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EDITOR,
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NASAL OBSTRUCTION AS A CAUSE OF HAY FEVER OR ASTHMA.

BY WM. WOODBURN, M. D., DES MOINES, IA.

I do not propose to treat this subject technically or theoretically, further than to simply say, I consider the first a mild form of the second, and both a reflex neurosis. I shall relate my experience and treatment of some half dozen cases illustrating my subject.

CASE I.—Mrs. H., a farmer's wife, æt. thirty-five, had been troubled for a number of years with hay fever from harvest time until frost came in the fall. Inspection of the nose in June, 1899, showed the lower right and both middle turbinated bodies greatly hypertrophied. Removal of the anterior and lower half of the middle and cauterization of the lower, gave complete immunity for the entire season. This patient could not, at any time, sweep the floor or ride behind horses against the wind, without violent paroxysms of sneezing, but since the operation has had no further trouble on this score.

CASE II.—Mr. D., æt. about thirty-five, a traveling man, the patient of our secretary, consulted me on August 28 last, in the midst of his annual attack of hay fever. Examination showed the entire nasal mucous membrane greatly engorged, as it always is during an attack. On the right side of the septum, near the floor, was a sharp septal spur, projecting at right angles about three-eighths of an inch, prodding the tumefied lower turbinated body. The removal of this spur under cocaine anæsthesia greatly modified the symptoms immediately, but a grateful frost, following in a few days, prevented an exact estimate of the benefit to be ascribed to the removal of this offending appendage.

This year, however, will furnish opportunity to determine how permanent the effect will be.

CASE III.—Wm. S., æt. four years, a great sufferer from asthma, at times when having a slight cold, to which he was very prone, to such an extent that he could not lie down for several days and nights. Relief had been sought in the higher altitudes of the Rockies and a residence of one year at Denver, but none came. I was consulted on November 25, 1899. An examination showed the post-nasal space almost occluded with adenoid vegetations. Of course I advised their removal, which advice was accepted, and their thorough removal, under the local application of cocaine, accomplished the purpose. This was the child of a brother practitioner, and in April this year I had a letter from the doctor, in which was the very gratifying sentence, "William has not had the asthma since you removed his adenoids, and is much better in every way.” This was especially pleasing since his suffering had always been more severe and constant during his previous winters.

CASE IV.—Male, æt. forty-five, Swede; occupation, bridge-builder. Had suffered annually for fifteen years with, first, hay fever, and, later in the season, asthma. In the summer of 1899 he anticipated his attack by a trip to the mountains of Colorado, where he found exemption and remained until the first frosts had appeared here, and then ventured to return. Immediately on arriving at Omaha on his return, his old antagonist met him and the battle again raged with even renewed vigor. He consulted me a few days after his arrival home, in a frame of mind ready to accept any suggestion which promised him relief. I examined the nasal passages. The right lower turbinated was greatly hypertrophied, and on the application of cocaine, 3 per cent. solution, the tumefaction largely disappeared and great immediate, but of course, temporary, relief was experienced. On the left side an immense septal spur on the osseous portion was found and removed. After the temporary swelling incident to the operation had subsided, his relief was, and remains, complete. I subsequently cauterized the enlarged right turbinated, since which time, he informs me, he has breathed more freely than he has done for fifteen years.

CASE V.—Lady, æt. about twenty-eight, married, had consulted all the physicians in her home town, except the one she should have at first consulted. Finally, in desperation she went to a young up-to-date homeopathic physician, who looked into her nose and assured her he knew what caused her trouble. He made an application of a solution of cocaine, and in a few minutes her respiration became nasal and normal. “Oh, what a relief! That is the first good breath I have had for weeks,” was her exclamation. Both lower turbinated bodies were immensely hypertrophied, and the doctor wisely advised their removal and referred her to me to do the work. When I first saw the patient she was a frail, pale, wan little woman, thoroughly exhausted. I saw her about six weeks subsequently, in a remarkably improved condition. Had gained about twenty pounds in weight and was plump, rosy-cheeked, and had entirely lost her distressed appearance.

The last two cases both occurred in the practice of my good friend, Dr. C. M. Harrington of Knoxville, Ia., and are all the more valuable because he is here to corroborate the fair statement of the cases I have made, and emphasize the beneficial results in a discussion of my paper.

CASE VI.—An old-school physician, æt. about thirty-five, had for a number of years been troubled during the late summer months and early fall with hay fever. Had sought relief from a number of specialists in his own school of medicine as well as prominent general practitioners. By chance he was directed to me, not knowing my school of practice, in the midst of attack in 1899. No satisfactory examination could be made until a 4 per cent. solution of cocaine had been applied and caused a lessening of the engorgement. There was plainly visible a large well-organized simple polypus hanging by a distinct pedicle from the right middle turbinated body. Its removal with the cold wire snare gave prompt and permanent relief and made a lasting and loyal friend of my old-school confrère.

In none of these cases do I mention any medication. None was used except to cleanse the mucous membrane of the viscid secretion always present in such conditions, and following such operations. All of the cases reported were mechanical obstructions, and demanded mechanical treatment, and no line of medicinal treatment would have done more than temporary good.

DETACHMENT OF THE RETINA—A CASE.

BY JAS. A. CAMPBELL, M. D., ST. LOUIS.

In May, 1885, Geo. H., age thirty-seven, came to me from Indiana, with the following history: He had been very nearsighted all his life, but had never worn glasses. His eyes had never troubled him in any other way, until a few days before he came. He then noticed a few floating white spots before his left eye. This gradually increased, and the vision of this eye, by degrees, grew less and less until after four days the sight of the left eye was gone. There was no pain in either eye.

Examination showed vision of the right eye was 4/200; with a — 13 Ds. glass, 15/200. With the left eye he could just distinguish light in the outer upper field of vision. With the ophthalmoscope only the lower fundus of the left eye could be made out, where the retinal vessels were seen up to the lower edge of the optic disk. The disk itself could not be made out, but above it a bulging, detached retina was prominent, with a hemorrhagic spot at its inner margin. In the upper outer fundus another separate bulging detachment of the retina could be plainly seen.

The right eye was highly myopic, with a myopic arching around the disk, and the entire fundus was mottled with small choroidal pigmentation spots, clearly of long standing.

The nature of his trouble was explained to him. He was kept quiet, and all forms of tobacco and stimulants were forbidden. He was placed on kali hyd. 3d, three times daily. In one week's time a remarkable change for the better had taken place. The detachment was much reduced. The wavy retinal vessels were seen climbing over its edges. The optic disk was visible. In two days more he could count fingers with the left eye at two feet. The improvement continued. On June 24 vision of the left eye was 15/200; right eye, 15/100. He then went home for a few days; the kali hyd. being kept up. July 13 he returned, saying that the improvement had gone on by slow degrees, until, suddenly, the vision of the left eye was again lost, on July 12. He was then placed on 5–grain doses of kali iod. 3d, three times daily, which was followed by very slow improvement, so that in three weeks he could again count fingers with the left eye at two feet, and the detachment, which had resumed its former dimensions, was somewhat reduced. I then went back to kali hyd. 3d, three times daily, which was again followed by improvement for some months after his return home. This, with some intercurrent medication, was kept up from time to time for a year, gradual improvement being reported until the sight of both eyes seemed about as it was in former years.

May 3, 1899, fourteen years after his first visit, he again came to me, reporting that his eyes had gotten along very well, with no particular trouble until in 1897, when he took a severe cold, which settled in his left eye, which became badly inflamed. He went to St. Louis, but did not find me, as it was during my summer vacation. Returning home, he consulted Dr. Knapp of Vincennes, Ind., who advised the removal of the left eye, as it seemed hopelessly involved by that time. This was done, and he progressed nicely, the vision of the right eye remaining about the same, though occasionally its vision seemed not quite so clear.

This was the situation until two weeks before his visit, when the sight of the eye began to grow dimmer. There was no pain present at any time, but vision gradually grew worse and worse.

Examination: V. R. = 8/200; where it had before been 15/100. Ophthalmoscope revealed some increase of the old choroidal atrophic mottling, with a red blurry optic disk.

He was placed on gels. θ, three times daily, for a few days, with evident improvement of the optic nerve congestion. He was then given kali hyd. 3d, four times daily. He returned home in a week, keeping up the same remedy. In three weeks he returned to me again, when examination showed decided improvement; with — 13 Ds., V. = 15/100 once more; still keeping up the kali hyd. 3d.; thus having been brought back to the condition which followed the treatment in 1885, and which had remained in *statu quo* for fourteen years.

Detachment of the retina is always a serious condition. It is not an unfrequent complication in high degrees of myopia. In three hundred cases collected in Horner's clinic, 48 per cent. were in myopic eyes. Its progress is generally unfavorable. It is usually treated by perfect quiet, rest of patients, after confining them to bed for some weeks, giving infusions of jaborandi and hypodermics of pilocarpin mur. Puncture through the sclera at the points of detachment, allowing escape of the fluid, has been advocated and performed by various well-known authorities, but has not been successful enough to ensure its general adoption. In rare cases spontaneous recovery has been observed, but I cannot think that the case here presented belongs to this class, for the original attack was in 1885, and was of such a degree that the vision of the left eye was reduced to mere perception of light in the outer upper field of vision. Under kali hyd. 3d, remarkable changes and rapid improvement took place. A relapse followed on his return home, after a couple of weeks. This again yielded to the same treatment, the vision of the left eye regained what it had lost, and remained in this condition for twelve years, when a severe general inflammation of the left eye necessitated its removal. Then, in a couple of years, the vision of the right eye became suddenly involved. Rapid improvement again followed the same remedy, and it was restored to its original condition. Hence, I cannot regard the improvement as either spontaneous or a coincidence, but think I am justified in attributing it to the direct result of the remedy given.

In the old school iodide of potash, in large and repeated doses, is a very common remedy in all intraocular diseases, and the more obscure the case, the more frequently and persistently it is used. Some cases are benefited by it, others not. The points I wish to make are, first to demonstrate the homeopathic possibilities of treatment in this serious disease; that when you have a homeopathic kali hydriodicum case, kali hyd. will probably help it, whether you give it in the third trituration or in more appreciable doses. In the case here reported kali hyd. 3d was certainly more potent than the 5–grain doses used for one week and then changed to the 3d trituration again.

CLINICAL CASES.

BY C. GURNEE FELLOWS, M. D., CHICAGO.

CASE I.—Early in 1899, Mrs. H. B., age forty-nine, presented herself for an opinion as to her condition. She had been hoarse for three or four weeks, and had a little inconvenience in swallowing, with no cough, but she complained of ordinary sore throat such as would follow an everyday cold. The main symptom was an excessive amount of mucus from the nose, nasopharynx, and pharynx.

Examination revealed a large pharyngeal ulcer on the left side, with enlarged cervical glands on the same side, and my suspicions were aroused as to its malignancy. Upon expressing such a fear the patient admitted that she had been examined by a surgeon who likewise had suspected carcinoma and advised its removal. She absolutely refused to think of operation at anybody's hands, and insisted upon my giving it the best treatment possible.

I cleansed it with the usual antiseptic solutions, and applied orthoform and other well-known and everyday methods for a week or two with a fair amount of relief, but the ulcer continuing, I applied specific treatment in the hope of clearing up the diagnosis, and, much to my delight as well as to the patient's comfort, the ulcer healed, the induration disappeared and the patient was, to all intents and purposes, well. After a number of weeks the external glands even diminished in size, so that I rather felt that the diagnosis of cancer was wrong and that it must be specific in character.

The patient ceased her visits, but returned in a few more weeks with a condition as at first, but upon the opposite side of the pharynx. Same treatment and everything else I could suggest did absolutely nothing for her relief, and I felt that the diagnosis this time must be carcinoma, but she refused even to have a small section taken for the purpose of diagnosis. Consultation agreed with me as to the malignancy of the growth, but operation was refused. She died in another month, practically from starvation.

I report this case after having read the article by A. Worrall Palmer upon cancer of the larynx, in the January number of this journal, because in the prelude, this method of diagnosis, and the application of this specific mixed treatment is advised, and because in this case it was followed by apparently successful results with what seemed to be an entire cure of the case, and therefore a clearing up of the diagnosis, but which, on the other hand, was followed by a return of the same condition, but upon the other side of the throat. The second point in the case is the great relief of all symptoms following the iodide of potash and merc. administered internally, and the administration of kali bi. and arsen., which most certainly had in the early part of the disease a very satisfactory effect. I believe that this case was cancer from the first, but that, contrary to expectations, it yielded beautifully to internal treatment.

Case II.—Mrs. M. E. C, age fifty-six, presented herself with a sensation of swelling in the throat accompanied by stinging pain, and with a history of from one to a dozen attacks of suffocation each twenty-four hours, much worse at night; otherwise no soreness of the throat or special sickness preceding these attacks, but they have been fairly constant for a year. She had become suspicious of the trouble being cancer, and, after having had some months of treatment from her family physician, she was more impressed with the fact than ever.

Examination revealed nothing in the way of foreign growth, but very much enlarged varicose veins at the base of the tongue, with a granular pharynx. Prognosis was favorable and the treatment as follows:

Glycerole of iodine to the pharynx and base of tongue, following cleansing and antiseptic solutions; galvano-cautery destroying the largest of the blood vessels, and moschus 3x internally. A complete cure resulted in less than thirty days; complete cessation of all attacks, which, of course, proved the diagnosis to be other than any malignant trouble, and, although the local treatment was probably efficacious, I believe that much of the trouble was of a neurotic type, incident to the climacteric, and that moschus deserves a good deal of credit for the result.

THREE KALI CARBONICUM CASES.

BY THOMAS M. STEWART, M. D., CINCINNATI, OHIO.

CASE I.—Patient, a tall thin woman; dark hair and eyes. Badly nourished as a result of mal-assimilation of food. Troubled with frequent attacks of styes on the upper right eyelid. Patient anæmic. Complained of frequent chilliness; chilly on least exposure. Physically and mentally patient was exhausted.

Some improvement was secured by correcting an eye trouble with glasses. Nux vomica, psorinum, and hepar of course acted indifferently. On a later visit the case was cleared up by the mention of the chilly sensation and the exhaustion. Kali carbonicum began an improvement and carried the case on to a point where diet did the rest.

The woman's means were limited, but she was able to carry out the diet direction, because her principal articles of diet had been meat and eggs. She was getting too much nitrogen. A generous supply of the carbo-hydrates; a direction to drink plenty of water, but not at meal times; and more exercise in the open air changed the conditions to healthful ones.

CASE II.—A young woman, well nourished, but not muscularly strong. Catches cold easily and is readily exhausted by muscular exertion. Sensation of a lump in the throat, with stitching sensation at each cold. With each cold has some cough, due largely to an elongated uvula. With each cold must "hawk" a great deal in the mornings to "clear the throat." The patient was a vocalist and suffered frequently from these acute colds and hoarseness.

The case had been prescribed for by several physicians. A study of the case brought out the kali carbonicum picture of "coryza with hoarseness; catches cold at least exposure to fresh air, and with each cold there is a stitching pain in the pharynx," and kali carbonicum 6x trituration cured the case, including the relaxed uvula. The patient has frequently presented this picture and each time kali carb. did the work.

Some additional benefit, in lessening the liability to these attacks, has been secured by the cold sponge bath each morning. Deep inhalation of fresh air three times a day, to aid in the oxidation of the food stuffs; and by inculcating the habit of daily attending to Nature's demands, whether there is any desire or urging in that direction or not.

CASE III.—Patient a nervous woman. Suffering from mixed astigmatism and pronounced insufficiency of the internal recti muscles, which we oculists denominate an exophoria. Patient suffered terribly from headaches, almost daily in their occurrence, frequently with nausea. The muscular trouble was cured by the use of prisms, the mixed astigmatism corrected by a glass, and there remained a severe backache. It was located in the small of the back as if there were a heavy weight pressing there; worse during menses, with bearing-down pain; patient was obliged to sit down frequently, on account of the ache. Her physician had prescribed sepia, cimicifuga, and natrum muriaticum—and in response to a question, “Could the eye treatment have had anything to do with apparently aggravating the backache?” I replied, “No; I think the relief of the headache has simply allowed the attention to be drawn to the backache.” I asked for other symptoms and one day received a little line stating that the “backache was worse after eating, and the patient could not walk much on account of the backache, was obliged to sit down frequently,” and kali carbonicum was advised. It cured the case.

ATROPHIC RHINITIS.

BY C. R. ARMSTRONG, M. D., THORNTOWN, IND.

Atrophic rhinitis is that chronic disease in which there is a wasting away of more or less of the mucous membrane, glands, and turbinated bones, and is generally accompanied by some abnormal conditions of pharynx and all the sinuses connected with the nasal cavities.

This is no new disease, but one physicians have had to deal with these many years; one we meet in practice every little while, and one which we cannot study too carefully, because the treatment for the disease in many cases ends in failure to cure.

I do not know that I will be able to say anything new of this morbid condition of nose, but will state a few things as I see them in practice day by day. There is always a favorable point about having a patient with this disease, along with the unfavorable ones. That is—the physician always has plenty of time to study his case and see every minute change in the recovery ere the patient is pronounced cured. This is more commonly known as ozena, or fetid catarrh, from the odor which accompanies the trouble. However, there is a form of the disease in which the atrophy is present, but has no fetor accompanying it. The latter is a much drier form with no secretions at all.

The ætiology of this disease has been discussed pretty thoroughly. It has been a question as to just what the initial symptoms and changes really are. In a majority of cases it is a secondary disease. That is, it follows other forms of rhinitis. Some authors claim that atrophic rhinitis follows the hypertrophic rhinitis. Others claim that there may be some shrinking in hypertrophic rhinitis, but that it does not end in atrophy, but comes from the purulent rhinitis: I believe they both are right. Either form I feel confident may precede atrophy. Again, I do not think that hypertrophic or purulent rhinitis is always followed by the atrophic form. If they were we would have many more cases of atrophic rhinitis to treat than we do at present. In my opinion this disease is brought on directly at times from various causes. Many a case of ozena has been brought on by the indiscriminate use of caustics on the mucous membrane of nose. Also injudicious cutting away of inferior and middle turbinated bones. Then again there is a predisposition to disease, especially in those people with a syphilitic and scrofulous diathesis. Excessive drinking of alcohol, excessive smoking of tobacco, working in poorly ventilated rooms or where there is a great deal of dust, and where there is an impoverished condition of blood from malaria or malnutrition, all have a tendency to set up this disease.

The name of this ailment tells much of the pathology of the disease. As atrophy implies, there is a wasting away of all tissues attacked. Upon examination the first thing observed is a dry, shriveled state of the mucous membrane of the nose and pharynx. The glands and follicles are all obliterated, which accounts for the dryness of the mucous membrane. The turbinated bones are dwindling away. Frequently the whole anterior portion of the turbinated bones is absorbed. This causes the nasal cavities to be so enlarged that we may see the pharyngeal walls from the anterior opening of nose. The glazed or dry appearance extends to the pharynx and in this manner affects the eustachian tubes. All over the nasal cavities and pharynx numerous granulations can be noticed. Tortuous and enlarged vessels run over the walls. As a rule there is not that bright red congested appearance of membrane as in other forms of catarrh.

Patients with this form of catarrh are frequently mistaken in diagnosing their own cases. I have had them to come in my office asking me to make a prescription for biliousness. They get that idea because they have a dry and coated tongue and a very bitter taste in mouth. After an examination is made you fail to find symptoms to corroborate the patient’s diagnosis, but will soon find the real cause. With the reflected light and nasal speculum it takes but a short time to satisfy your mind from conditions of nose that have all the symptoms of ozena. There is a discharge made up of mucus which is very thick, therefore not very easily expelled, and as a result finds its way into all the fossæ and crevices in the nose. It is not long until this is dried into crusts which obstruct the passages of air, and being retained, decompose, throwing off a peculiar, penetrating stench. These crusts adhere very firmly to the membranes. The patients will remove them by artificial means, owing to the uncomfortable feeling produced by them. When the scabs are torn away there may be an oozing of blood. A stuffed up and oppressed fullness in the superior and posterior portion of the nasal passages is present. In the first stages of the disease the mucus will fall down from the palate in small slugs or masses, which as the disease goes on become more and more tenacious and more of a muco-purulent nature. In the beginning this discharge can be “hawked up,” but soon it becomes too thick and dry. While the membranes may be so irritated that there will be a free discharge of blood, still there is no real ulcerative process. The septum in some cases is perforated, but this is caused more by tearing away the dried-up discharge than anything else.

The sufferers from ozena are never the strong and vigorous people. They are generally anæmic and having family histories which would make a physician think that the diseases were hereditary. These discharges being retained so long the poison may be absorbed into the blood, and soon the whole system will show the effects of the poison. In children the nostrils are so filled up that they can scarcely breathe at night, and it will not be far in the future when the child will be weak, nervous, irritable, and unable to sleep well. The stomach raises a disturbance as the disease gets older, which is accompanied by an occipital headache. The food which is eaten goes for naught, because the system does not seem to get the desired nourishment. Taste is destroyed, appetite gone, loss of energy for everything is apparent.

The patients scarcely ever can detect the bad odor unless their attention is called to it. After being told a few times about the odor of the breath they will shun public gatherings. If the patient is a woman, who because of offensive breath is barred from society, she will become morbid and hypochondriacal in time. There is so much of the thick mucus hanging on the walls of the pharynx that the openings into the eustachian tubes are filled up and in a short time a certain degree of deafness appears—roaring in the head and other manifestations of ear trouble. As soon as the hearing is noticed to be abnormal the patient will be ready to consult some physician.

With these symptoms it may not require much time to make a correct diagnosis, but it may be some time ere the patient is entirely free from the trouble even if he does use good homeopathic treatment. Then it is the treatment which interests us most. The patient at the beginning will ask if you can cure him and how long it will require to do it. The physician necessarily must guard his prognosis, especially if it is a case of long standing. If there is much atrophy, which has extended over several years, a permanent cure is very doubtful. But even with these cases much can be done to make patients more comfortable. Correct the odor, the dryness, and the formation of scabs. If the case is not of too long standing, very likely you will be able to produce a healthy condition of the mucous membrane. If you are so fortunate as to produce a cure, the patient will always remember you for it, and you will or should feel proud of it yourself. Much will depend on occupation, age, and persistence with which the patient carries out treatment.

The treatment, to be beneficial, implies the discovery and removal of all predisposing and exciting causes. To do this will require both local and systemic treatment. No cure can result unless good constitutional treatment is persisted in. When taking the case it is wise to inform the patient that he must expect treatment through several months, and even then the case must be examined once in a while or there may be a recurrence of the disease.

Too much time cannot be spent in a careful examination of the patient. Be certain the cause of trouble is ferreted out. The course of treatment will depend upon the cause of the disease. After thorough examination a course of treatment is planned. As I have said before, each case must be studied. There are no specifics for the disease.

In the local treatment the important object is cleanliness. The mucous membrane must be kept in a perfectly clean condition all the time. This is the main object of all local treatment. In some cases I assist nature to heal parts by getting a slight stimulating effect of medicine.

It is not always an easy task to remove all the dry crusts, but where the scabs are very dry I use an application of peroxide of hydrogen on cotton, or with the atomizer, to soften them. When the atomizer or douches are used post-nasal injections must be given as well as through the anterior chambers of nose. Any application can be used which will soften up scabs. Can use "Dobell's Solution," solution of sea salt, listerine, or glycerine. After all the crusts have been removed, others must be prevented from forming. This I do by keeping on an application of glycerine. A very good formula to keep the nostrils free is calendula and glycerine, at 2 drams to ounce water, and used in nebulizer or directly applied on cotton. When the odor is present after removal of scabs, I use permanganate of potash, 10 grs. to ounce in spray, or aristol in lavolene used in nebulizer.

After the cleaning process has been gone through with and all the mucous membrane is perfectly clean, naturally it is ready for some healing application. A good one to use is calendula and hamamelis in lavolene. If there should be any ulceration of septum, apply an ointment of yellow oxide of mercury, 10 grs. to the ounce. This will heal ulcer in short time.

Where membranes need some stimulation a glycerite of tar, hydrastis, or eucalyptol in nebulizer will be found to be of service.

Many times patient will complain more of the deafness than anything else. When you have this complication it will be necessary to give attention to some special treatment for the pharynx and eustachian tubes. The latter must be kept open by Valsalva’s method or the Politzer air bag.

In selecting the internal remedy keep in mind the constitutional and local lesions. Often I use the internal remedy locally; say 5 to 20 drops of tincture to ounce water. Make yourself confident that you have the indicated remedy. There are many remedies which are of service. Some of the more common ones, which have syphilitic taint are, aurum, kali iod., mercury, nitric acid, argentum nitricum, and calc. iod. In scrofulous diathesis and ill-nourished patients such remedies as aurum mur., silicea, calc. phos., sulph., phosphorus, ars., hepar sulph., alumin., kali bich., cal. carb., and graphites are useful.

All through the treatment the physician should have perfect control of patient. Should be able to direct his diet and hygiene. Use all means that will recuperate the general health. If patient is laboring day by day in dust and dirt, he may be compelled to change his occupation.

It is only by looking after the general health that the physician may expect to be rewarded with any success.

SPRAYS.[1]

BY FRED D. LEWIS, M. D., BUFFALO, N. Y.

In considering the subject of sprays, it is not my intention to present to you a number of formulas that I have found useful in my practice, but to consider the matter on a broader and more general basis. That sprays have been, and are still used, in various conditions with the most gratifying results, we all know. But that they should be prescribed to a much larger extent than they now are is a fact that the physician, as a rule, is not aware of.

We have learned to know that the skin is one of the great vital organs of the human system. That if its action is impeded, the kidneys and intestines are thereby given a greater amount of work to perform. That with the morning sponge, followed by a brisk friction and an occasional Russian or Turkish bath, in chronic cases, such as rheumatism, we can expect quicker and better results from our remedies.

The public generally have been educated to that point where they recognize the importance of proper care of the teeth. They not only regularly cleanse them, but at stated intervals, usually every six months, go to the dentist and have a thorough examination to anticipate rather than wait for trouble.

Many persons have learned that a lavage of the stomach, in the shape of a cup of hot water, before meals, has converted a sluggish digestion into a normal one.

We are all familiar with the structure and object of the nasal cavities. The tortuous turbinateds provide a large surface for the air to secure heat and moisture, before reaching the lungs; and also remove from the air such impurities as are of a solid nature. Now we all know that the atmosphere of cities, especially where there are large manufacturing interests, is loaded with impurities, such as soot, dust, particles of pavement ground to impalpable powder, etc., etc. This fact can easily be demonstrated when the city is on a plain or in the neighborhood of a large body of water. When in the city the air seems pure, the sky unobstructed, and no evidence of floating particles of matter, if an observation is taken from a few miles' distance, the city appears to be encompassed by a cloud.

That the disposition of foreign matter on the sensitive lining membranes of the nose should produce disturbances, there can be no doubt.

The only point I wish to bring out, and I hope it may stimulate some discussion, is this: Should not the care of the nasal mucous membranes be considered as important as the care of the skin and teeth?

In recent years I have asserted to my patients that the spray, in my opinion, is as essential on the toilet table as the toothbrush. As to the nature of the spray to be used, I think one must be guided by conditions. If there has already been a catarrhal condition established, then some remedial agent had better be employed; but if used simply as a prophylactic, then a neutral cleansing solution would be preferable.

I think this subject is deserving of profound consideration, when we know that there are establishments in most of our leading cities that advertise the cure of catarrh for so much a month. Their methods are simply to insist on the patient coming to their offices daily, and having their noses thoroughly cleansed. And they are curing many cases. Would it not be wise to educate our patients, not only to keep their own noses clean, and thus cure themselves, but, by attending to themselves early enough, avoid the development of that, perhaps, most prevalent of all diseases, catarrh?

GALVANISM IN NASAL HYPERTROPHY.[2]

BY JOHN B. GARRISON, M. D., NEW YORK.

Hypertrophic rhinitis is one of the most frequent of the diseased conditions pertaining to the nasal cavities that we are called upon to treat, and the question of the most suitable method of treatment is to be decided with care.

We have all used, for the removal of the excess of tissue, perhaps, with more or less success, the acids, the actual cautery, or some form of cutting instrument, but the patient, at least, will welcome a method that promises a good result with the least amount of pain at the time of treatment, and the least soreness afterward.

I have found that the application of the galvanic current does, in many cases, furnish just the method desired, and I shall beg your attention for a few minutes while I speak of the method as I practice it.

I shall not burden you with my ideas of what cause most enters into the production of these nasal hypertrophies, leaving to you the perusal of the text-books that will give all the knowledge extant upon the subject. We do find an increase of the nutritive forces, and our treatment must be directed to a lessening of the blood supply in some way. Of course where there is a local source of irritation, that must be removed at once. If it is a deflected septum that is causing an irritation by contact with the opposite side, suitable means must be adopted for its repair before attempting to treat the hypertrophies opposing the irregularities of the septum.

The hypertrophies that I shall speak of as being most amenable to treatment by means of the aid suggested in my title are mainly those of the turbinated bodies: and, of these, the inferior is the one most often enlarged.

It may be confined to either extremity, or the whole body may be the subject of hypertrophy. When, as is sometimes the case, the bony portion of the turbinate has become enlarged, the saw, and not electricity, will be the best means of cure.

But when the occlusion of the nares is caused by true increase of tissue we have, in galvanic electricity, a potent agent to safely and rapidly remove the obstruction.

To prepare a case for treatment, I always first thoroughly irrigate the nasal cavities with some antiseptic fluid, using the post-nasal syringe. The solution that I most frequently use is Electrozone one part, and tepid water four parts. Then an application of a four per cent. solution of cocaine is made to the location about to be treated, simply to prevent the little pain which accompanies the introduction of the electrode.

The electrode I use is a slender needle about the size of an ordinary darning needle, of suitable length for easy use on the part selected, and I insulate it by dipping it in shellac and laying it away until it is perfectly dry, then scraping away the insulation as far from the point as it is calculated it will be impaled into the tissues. It is fastened into an ordinary needle-holder and connected with the negative pole of the battery, when it is introduced into the tissue at the point selected. The patient is then given the sponge electrode connected with the positive pole of the battery and is told to grasp it firmly, and the current is slowly turned on until the meter registers from three to five ma., which current is allowed to remain stationary for about five minutes, unless the patient is very nervous, when three minutes should be the limit.

The current is now turned off as gradually as it was turned on and the needle carefully removed. I do not attempt a second treatment at the same point until a week has expired, and in some cases two weeks can be permitted to go by before the shrinkage due to the electrolysis has subsided. The stronger currents have been tried, but the strength I have used and given here acts much more pleasantly and gives equally good results.

During the summer just past I had the opportunity of noticing the reduction of an enormously hypertrophied inferior turbinate in a most unexpected manner, which I am glad to relate at this time.

A lady of about fifty years of age, who was stopping at the hotel at which my family and myself were located, came to me one day to ask my opinion as to her eye and nose. She had had a stricture of the nasal duct for a number of years, which had been duly dilated several times, and for a considerable time had had a dacryocystitis which annoyed her greatly, and from which she was able to press a large amount of mucus and pus from the canaliculus.

The inferior turbinate on the affected side was hypertrophied for nearly its whole length and was in contact with the septum for some distance at the anterior extremity, being of a deep red color and very sensitive to touch. I told her that I believed it would be necessary to remove the turbinate with the saw and advised its removal as soon as possible, giving it as my opinion that it would be necessary to do the operation before the condition of the eye could be relieved. The patient admitted the force of my argument, but was inclined to wait a while until she could get her courage up a little higher. Meanwhile she wanted the canal dilated and begged me to do it. Visiting New York, I supplied myself with a canaliculus syringe and a set of Bowman’s probes, and on my return announced myself ready to commence treatment. I proceeded to insulate the probes in the manner alluded to for needles in nasal work, scraping the points bright for about a quarter of an inch.

Before introducing the probe I washed the sac out thoroughly with a fifty per cent. solution of enzymol, and then, connecting the probe electrode with the negative cord of a galvanic battery by means of an artery forceps, introduced it (No. 2,—a No. 1 would not pass in the ordinary manner with considerable pressure) into the canal, and turned on the current until the meter registered two milliamperes. Using just enough pressure to guide the electrode, it gradually found its way along the canal, and in less than five minutes it had entered the nasal cavity without causing the loss of a drop of blood. In three days I passed a No. 4 in the same manner, and four days later a No. 6 passed easily. Three days after this a No. 7 was passed, and that size was passed three or four times afterwards at intervals three or four days. After the first passage of the No. 7, all of the solution used for the purpose of cleansing the sac passed through into the nasal cavity directly from the syringe, and there was no further collection of pus in the sac during a week in which the syringe was not used.

The point I wanted to bring out, however, is that after the second treatment by electricity, the color of the mucous membrane covering the turbinate began to grow paler, and at the end of the treatments the entire body had contracted sufficiently to permit free and easy drainage and natural respiration.

I am led, by this, to the thought that it may be good treatment in many cases of hypertrophy of the inferior turbinate, and possibly the others as well, to use the insulated probe electrodes in the lachrymal canal with the weak current, not exceeding one or two milliamperes. The careful use of this method may prove it to be a valuable addition to the present means of treating a class of cases that are troublesome to the patient and the doctor.

THE PATHOGENIC AND THERAPEUTIC ACTION OF RHUS TOX. UPON THE EYE.

BY CHARLES DEADY, M. D.

In the collection of provings of rhus toxicodendron made by Samuel Hahnemann, and published in Vol. II. of the Materia Medica Pura, the number of symptoms relating to the eye and its neighborhood is so large as to lead to the supposition that, if the theory of homeopathy be a correct one, this drug should prove of special value in the treatment of diseases of the visual organs. Never did an hypothesis receive better support when reduced to practice; and if the efficacy of the law of *similia similibus curentur* were compelled to rest upon a single test to demonstrate its truth, few better selections could be made than that of rhus tox. in the department of ophthalmology.

Many of the symptoms contained in the Materia Medica Pura are indefinite, and the great majority of them point to apparently superficial diseases, but when we consider the fact that at the time these provings were made the science of ophthalmology, as understood at the present day, actually had no existence, that the principle of the ophthalmoscope had not yet been discovered, and that the methods of precision in the examination and diagnosis of diseases of the eye now available were at that time unknown, this is little to be wondered at. And we are compelled to admire the industry and energy of the men, some of them of our own day and generation, whose tireless labor has sifted and arranged the numerous symptoms of this and other drugs and indicated the method of their proper application to the various pathological processes.

When the New York Ophthalmic Hospital was placed in charge of the adherents of the homeopathic school, they were confronted by the fact that no definite materia medica of diseases of the eye and ear, as such, was in existence, and in order to ascertain the remedy for a given case of disease they were obliged to take the conditions throughout the body, and by comparing these with the general materia medica find a suitable drug for the totality of the symptoms. Had they been content with simply curing their cases in this routine way little would have been gained, but they made it a rule to take down the special eye or ear symptoms in each case with great care, and when a drug had cured a certain case of disease the eye or ear symptoms which had disappeared under its use were carefully noted. With a multiplicity of cases, and a systematic verification of symptoms, a valuable special materia medica of these diseases was compiled, and the curative properties of drugs in the various pathological entities of the eye and ear were definitely demonstrated and their characteristic symptoms for each disease mapped out.

Under this methodical procedure the relative value of drugs apparently indicated in these diseases gradually became better known; some, although presenting many and varied symptoms, were found by experience to be superficial and evanescent in their action, while others proved of the greatest efficacy in the most serious lesions, and became indispensable in the armamentarium of the physicians of the hospital staff.

In the latter group, rhus tox. speedily assumed prominence as a drug of special value in ocular disease, and this was enhanced by such a large measure of success in its application over a wide range of affections that it came to be regarded (at least in this hospital) as a veritable sheet-anchor in ophthalmological work, and as time passed it was used more or less in almost all the acute diseases to which the eye and its adnexa are subject.

A prominent symptom of rhus is great swelling of the eyelids. This it has in common with a number of other remedies, but differentiation becomes less difficult when we remember that rhus is specially indicated when swelling and œdema of the lids are the result of the deeper and more serious lesions. After the operation for cataract, one of the first symptoms indicating danger is œdematous swelling of the lids, and no drug in the materia medica compares with rhus for insuring the safety of the eye. When the pathological process becomes advanced, even to the appearance of pus within the eyeball, still we may confidently rely on this remedy, which has cured many such cases when they were apparently hopeless. In all postoperative complications it is of the greatest value, and too much emphasis cannot be placed upon this statement.

When a sound eye takes on sympathetic irritation from its diseased fellow, the fact is first manifested by a certain amount of swelling of the lids and more or less profuse lachrymation, the latter another valuable indication for rhus tox. In this condition I have personally used it many times with complete success, and it is the first drug to be thought of in this extremely dangerous complication.

It is a well-known fact at the present time that rhus is particularly applicable in rheumatic conditions, especially where these are resultant upon a wetting or exposure to dampness. This, together with its nightly aggravation, points to another sphere of usefulness in rheumatic iritis, where it will prove all-sufficient when the characteristic symptoms exist. In suppurative iritis and cyclitis it is very serviceable, no matter what the cause.

The symptom “while he turns the eye or it is pressed, the eyeball is painful, can hardly move it,” indicates its use in acute retrobulbar neuritis, which causes this symptom exactly and is well known to be frequently due to a rheumatic diathesis. The same symptom may call for its use in tenonitis, in which the stiffness, difficulty of and pain on moving the eye are specially prominent and which also has swelling and œdema of the upper lid, chemosis of the conjunctiva, and protrusion of the eyeball; all symptoms of rhus tox. The idiopathic form of this disease is almost always rheumatic or gouty in origin, furnishing still another indication for the remedy. In orbital cellulitis we have swelling of the lids, chemosis, protrusion of the eyeball, almost complete abolition of motion with pain on the attempt and also on pressure, aching in and around the eye, with the probable formation of pus in the deeper structures, all conditions curable by rhus tox., which is one of the best remedies for this disease whatever may be its origin, traumatic or otherwise, and has cured many of the most desperate cases.

In panophthalmitis, or suppurative inflammation of the eyeball, we find the swollen lids, difficulty of, and pain on motion, chemosis of the conjunctiva, severe pain in the eyeball, lachrymation, etc., again indicating the remedy. Rhus tox. is one of the few drugs that have cured this most fatal of lesions, and its success in restoring the integrity of the eye, in some cases where this result has seemed almost impossible, is a matter of record.

The symptom “heaviness and stiffness of the eyelids, like paralysis, as if difficult to move the eyelids,” would seem to indicate its use in ptosis, and this condition as well as paralysis of certain of the ocular muscles, is curable by rhus tox., especially if due to wetting or dampness. Such cures have been made frequently in the clinics of the Ophthalmic Hospital. Erysipelas of the eyelids often presents the characteristic symptoms of rhus. Of course the swelling of the lids is always present, but many of these cases have in addition the chemosis, hot lachrymation, the characteristic pains and aggravation, restlessness, vesicular eruptions, etc., and it is a valuable and efficient remedy when these exist.

Although rhus tox. is specially useful in the most serious inflammations of the deeper and more important structures of the eyeball and surrounding tissues, its sphere is not confined to these conditions alone, but seems to cover almost all the acute diseases to which the visual organs are subject. Given a rheumatic origin, especially if it be from exposure to damp or wet weather, with profuse lachrymation (pain in and about the eye) a tendency to chemosis of the conjunctiva, œdema of the lids, photophobia, and the characteristic aggravation and restlessness at night, and this valuable drug will rarely be found wanting in any of the inflammations of the conjunctiva, cornea, or lids.

I have many times cured with it acute catarrhal conjunctivitis, phlyctenular conjunctivitis and keratitis and ulcers of the cornea, and have subdued the acute aggravations of conjunctivitis trachomatosa, where the above symptoms, or some of them, were present. It is also frequently successful in the treatment of abscess of the lid, which, while not a serious, is an extremely painful disease.

Dr. W. A. Phillips, in an article published in the *Jour. of Oph., Otol. and Lar.*, July, 1899, page 224, recommends the use of rhus tox., “when the ciliary muscle itself seems to be the special seat of trouble; when its muscular tone is disturbed from previous straining, and when inability is present after using the eyes for reading any considerable time, notwithstanding optical correction.”

He has had much success with the drug in these cases and considers that its action here is on a plane with that on lameness or soreness due to rheumatism. In my opinion another factor may be spoken of. One of the differential points between arsenic and rhus is that the arsenic patient is *actually* so weak that he cannot do what he would wish, while the rhus patient *feels* so weak that he cannot do it, but by making the effort he can overcome his weakness and accomplish what he desires. This seems to indicate in the rhus case an indisposition to exertion due to want of *tone* of the muscular system, and this explanation applied to the ciliary muscle would account for the successful action of this drug in the class of cases indicated.

I would not have it understood that I consider rhus tox. an universal panacea for all the inflammatory diseases of the eye; all of these affections are many times extremely variable in their presenting symptoms and other remedies are frequently called for, but the drug under consideration is one of the first importance and is most reliable and efficient when accurately prescribed.

TREATMENT OF SARCOMA WITH THE MIXED TOXINS OF ERYSIPELAS AND BACILLUS PRODIGIOSUS.

BY A. WORRALL PALMER, M. D., NEW YORK.

The numerous modes of treating sarcoma or any other variety of cancer, and the constant experimentation on the part of the profession with new methods, only go to show how inadequate is our ability to meet this intractable disease.

These neoplasms are not so rare, as there are ninety-nine authentically recorded cases, situated within the restricted domain of the naso-pharynx and pharynx.

For these reasons, and because I have been able to find only one case of sarcoma treated with Coley's fluid reported in our homeopathic literature, do I take the liberty of occupying your time with the *résumé* of my investigations into the subject and my meager practical experience.

Although surgery is, at present, the best method to meet this condition, personally I believe that more investigation into or trials of the remedial treatment should be made, because cancer is a constitutional disease, and it so very frequently recurs after removal with the knife.

Apropos to this, C. Mansell Moullin says in the Boston *Medical Journal*: "There is at least as much hope after an internal remedy that causes disappearance by atrophy or fatty degeneration as from the most extensive removal by operation. On *a priori* grounds there may be even more."

Among the numerous drugs or substances which have been experimented with are the interstitial injection of alcohol 40 per cent., by Haase; the injection of Pure Yeast Ferment, by De Bracher; subcutaneous use of 50 per cent. solution of the fluid extract of chelidonium majus re-enforced by same drug per orem; the cataphoric diffusion of mercury from gold electrodes used by Massey; and lastly the mixed toxins of the streptococcus erysipelas and bacillus prodigiosus.

From my research the last is the only one that has attained any success or wide reputation and not been relegated to the usual oblivion of other medical fads. The reason for this I consider to be because Dr. Coley has not only been persevering, but scientific, unbiased, and very cautious in its advocacy. At first he hoped and believed that in some form it would be beneficial in all forms of cancer; but he now only recommends it in sarcoma, and claims marked results only in the spindle-celled variety of this.

As in many other cases, the discovery of the influence of erysipelas on sarcomatous growths was by investigation founded upon accidental occurrences, to wit: Busch reported a case of multiple sarcoma of the face cured by an attack of facial erysipelas; Durante, a sarcoma of the neck; Biedert, an enormous round-celled sarcoma, including the mouth, nose, and pharynx; Bruns, a melanotic sarcoma of the breast; Gerster and Bull, each a recurrent sarcoma of the neck; all cured or disappeared with no return, after an erysipelatous attack. This happy result does not always follow erysipelas, as cases of sarcoma relieved by erysipelas, and later recurring or progressing after the attack is over, are reported by Busch, Nelaton, Deleus, Richochon, Winslow, Powes, and Dowd.

On account of these accidental cures a few observers produced erysipelas artificially by infusion with the living culture, with success in many cases.

Then almost simultaneously Lassar of Berlin, Spronck of Utrecht, and Coley of New York, believing that the curative action of erysipelas lay in the toxin of the living culture, experimented and found that they could produce equally good results with toxin, thereby avoiding both the danger and discomfort of the patient passing through an attack of erysipelas.

It has been shown by different observers that the combination of certain bacilli with disease toxins makes such toxins more potent, and Rogers of Paris demonstrated that the combination of the bacillus prodigiosus with the streptococcus of erysipelas greatly augmented the virulence of the streptococcus on rabbits. Thereupon Dr. Coley used the combination on the human subject in sarcoma with far better results than before.

Regarding this, Dr. Coley says he cannot say exactly what part the bacillus prodigiosus plays in the cure of sarcoma, but remarks that the only cases cured were treated by the combination.

This preparation, the combined toxins, had been given the name of Coley's fluid, and that used during the last seven years has been made by Dr. B. H. Buxton of Loomis Laboratory.

Until about five years ago the toxins were made from cultures from a fatal case of erysipelas, but since that, sufficient strength has been obtained by passing the cultures through about fifty rabbits. The method of the preparation is virtually this: the mixed unfiltered toxins of the streptococcus of erysipelas and the bacillus prodigiosus are made from cultures grown together in the same bouillon and sterilized by heating to 58 degrees C. and then diluted in a sterilized menstruum.

In a recent conversation with Dr. Buxton he said that at present he made a double sterilization and then added some drugs such as thymol to preserve the preparation.

Dr. Coley, in his exhaustive article in the *Jour. Am. Med. Assoc.*, August 20 and 27, 1898, affixed a table of fifty-seven cases of cancerous tumors treated with either his fluid or other preparation of erysipelatous poison with cure, or at least disappearance of the then present manifestation of the disease and lengthening of the usual period of a recurrence of the condition.

The following is a list of cases of sarcoma of the nose and throat treated by cultures of erysipelas, or Coley's fluid, the physicians in charge, and the time the patient is living after treatment at the time of the report in Dr. Coley's paper, in 1898:

(a) A spindle-celled sarcoma of the neck and tonsils, inoculated culture—patient living six years after.

(b) A spindle-celled sarcoma of the parotid; it had been extirpated twice previous to treatment—patient living one year after.

(c) A sarcoma (mixed celled) of the parotid—patient living three years after. The foregoing under Dr. Coley's care.

(d) A spindle-celled sarcoma of the palate and pharynx extending to the vocal cords—Dr. W. B. Johnson—living four and three-quarter years.

(e) A round-celled sarcoma of antrum, pharynx, and neck—Dr. L. L. McArthur—child aged five years, weight gained from 37 to 69 pounds—later, fatal recurrence.

(f) A round-celled sarcoma of parotid, size of the fist—Czerny of Heidelberg—living over a year.

(g) A spindle-celled sarcoma of the parotid—Horace Packard—living two and three-quarter years.

(h) A round-celled sarcoma of the neck—H. Montague—slight return in six months.

(i) A recurrent sarcoma of the neck and tonsil—J. O. Roe—six months after treatment died of erysipelas.

The mode of administration is cumulative. The injection is of course to be made under the most thorough antiseptic principles attainable. It is by far preferable to make the injection into the growth itself, although, if this is impossible, it may be introduced into the nearest accessible point, but in the latter case the dosage needs to be doubled.

As a rule one-half drop is the initial dose, and this is increased one-half drop each succeeding day until toleration is reached. This is evidenced by the natural reactionary fever rising to 102° or 103° F. In such case the following dose should be the same as the preceding, and if it should again go so high reduce the next dose one-half drop. The dose is increased in this manner until the maximum is attained. When applied to the neoplasm itself 8 drops is the full dose, or if elsewhere, double that amount, 16 drops.

This last amount is to be continued daily until the tumor has disappeared.

The toxin may commence to reduce the tumor in a week, but its administration should not be abandoned in less than three weeks' trial. The time necessary to effect a cure is very variable; occasionally the neoplasm will almost disappear in two weeks, while on the other hand it may take several months.

The reactionary symptoms are a chill, followed by fever, generally lasting about three hours, although occasionally it may continue twelve hours; acute transitory swelling of tissues in the immediate vicinity of injection; usually myalgic pains commencing at point of injection and radiating frequently over the whole body; in the more severe reactions there is nausea or even vomiting—in my own case it produced a weakening menorrhagia.

CASE.—Mrs. E. C., æt. thirty-four years. A tall, thin woman of neurotic temperament.

Family History.—Father had chronic bronchitis, but died of kidney disease. Mother was an invalid for seven years with rheumatism of hip and knee until death, which was caused by apoplexy; a sister died of gastric disease. The patient married eleven years; has two children living; boy at nine months died of entero-colitis; boy three and one-half years died of fall from window; two miscarriages. At ten years æt. the patient had diphtheria; at twenty-six, pleurisy; at thirty-one years, rheumatism of left shoulder and post-cervical region. It is impossible to obtain any indication of hereditary predisposition.

Subjective Symptoms.—Complains of post-nasal dropping of mucus, constant short hacking cough, malodorous breath, pain in region of spleen; aggravated when lying down and throbbing in character when walking rapidly. After discovering the swelling in the throat and speaking of it she admitted there had been a sensation of a lump in the throat for about a year, but so slight she considered it of little consequence.

Objective Symptoms.—Nares: Rhinitis sicca, covered with dry crusts, but turbinated bodies hypertrophied.

Naso-pharynx and pharynx: Mucosa slightly hyperæmic, follicles inflamed and enlarged. On the left side of these cavities is a sessile swelling, the general surface of which is much inflamed, and half of the surface is covered with varicose veins about one-eighth of an inch in diameter; it extends more than half the width of the pharynx and vertically from the vault above to the lateral sinuses below; is neither painful nor hyperæsthetic; it has a boggy feel, but not as soft as an abscess. The tumor springs from the posterior wall of the pharynx, not connected with the tonsil, as the left posterior pillar lies in front of the neoplasm and can be lifted free from it. Neither of the tonsils is inflamed nor hypertrophied; a few cervical lymphatics on the left side are slightly indurated, but slightly sensitive—if at all.

The swelling had probably existed longer than an abscess would be in forming, and there was neither pain nor fluctuation. Still an exploratory incision was made, but with the expected negative results.

Although the tumor was situated over the principal chain in lymphatics of the pharynx, it was not nodular, but smooth. Therefore the neoplasm was probably not of lymphatic origin, but an implication of the muscular tissue behind the pharynx.

A specimen was submitted by Dr. Klotz, the pathologist of the hospital, and the provisional diagnosis of angio-sarcoma made—sarcoma because it seemed to spring from the muscular tissue and apparent predominance of blood-vessels, and of the angiomatous variety because of the enlarged blood vessels on the surface.

The removal of the specimen for microscopical examination caused quite a severe hemorrhage, lasting about two hours, notwithstanding the employment of the usual hemostatics.

The microscopist pronounced it a small round-celled sarcoma.

I showed the case to the Academy of Pathological Science, where two general surgeons who examined the case advised against extirpation of the tumor, because of its close proximity to the important blood vessels and nerves of the neck, an opinion I entirely coincided with, because of seeing two similar cases before. This agreement decided me in determining to try the mixed toxins as the treatment promising the best results for the patient.

April 4. Commenced injections with one-quarter of a drop. I diminished the initial dose one-half because Dr. Coley personally advised it, as he thought the possible reactionary local swelling might seriously interfere with respiration.

April 14. The dose was increased one-quarter drop each day to date—when she took only two drops, because it was deemed advisable to omit treatment two days during menstruation on account of great weakness of patient.

April 20. Increased dose half drop per diem—on 16th and 19th treatment omitted on account of debility—dose 4 drops, which dose was continued till April 23, when on account of the temperature twice having risen to 103° F. and menorrhagia having supervened only ten days after previous regular menstruation, I thought it prudent to reduce dosage to 3½ drops, which was continued until April 26. Examination of pharynx to-day for first time showed a decided diminution in the congested appearance and size of the tumor. Formerly the tumor pushed the posterior pillar forward, so that, if the pillar could not have been lifted away from swelling by the ring probe, it would have seemed to be part of it; while to-day a small space could be distinguished between the tumor and the pillar. Dosage 4 drops.

In *résumé*, I would call attention to the apparent susceptibility of the patient to the toxin. Because, although she never received over half the maximum dose, the following reactionary symptoms developed: Of the seventeen days on which full records were kept, on thirteen she had chills after every dose; there were muscular pains throughout the left side, occasionally extending to the right—one-third of the time the patient was nauseated, and three times vomited—the average temperature was 100.8° F.; twice it did not rise at all after injections of ½ or 2½ drops. ’Tis well to bear in mind that chills very seldom occur after the third injection.

Finally, I wish to thank Dr. Clausen, resident physician, who carried out most of the treatment while the patient was at the Ophthalmic Hospital; also Dr. Bernard Clausen, who continued it after she returned home.

REPORT ON “HENPUYE” IN THE GOLD COAST COLONY.[3]

BY ALBERT J. CHALMERS, M. D., VICT., F. R. C. S. ENG.

Henpuye, or dog nose, is a disease frequently met with in the Gold Coast Colony and in certain portions of its Hinterland. The hideous deformity of the face which it causes is very striking to anyone who has lived in this part of West Africa. It is also known on the French Ivory Coast under the name of “goundu” or “anakhre,” but “henpuye” is the native name (Appolonian) for the disease on the Gold Coast. The peculiar nature of the disease and the fact that, as far as I could find, very little was known as to its nature led me to make the inquiries which are now embodied in this report. I regret very much that I am unable to refer to original papers on the subject or to be certain that I have the full literature, but my excuse is that libraries do not exist in West Africa. The only references which I have met with are those mentioned in Dr. Patrick Manson’s work on “Tropical Diseases” (p. 594), and they are those of (1) Professor Alexander Macalister (Royal Irish Academy, 1882), (2) Surgeon J. J. Lamprey, A. M. S. (*Brit. Med. Jour.*, vol. ii., 1887), (3) Dr. Henry Strachan (*Brit. Med. Jour.*, vol. i., 1894), and (4) Dr. Maclaud (Archives de Médecine Navale, 1895). It is by the kind permission of the Governor of this colony, Sir Frederick Hodgson, K. C. M. G., that I am allowed to publish this report. I am much indebted to Captain Armitage for his kindness in giving me information with regard to the different places in which he has noticed this disease in his travels, for drawing my attention to notes of the late Mr. Ferguson on the presence of the disease in Akim and Kwahu, and for making a painting of an advanced case of the disease; also to Dr. Henderson, the chief medical officer of the colony for many kind suggestions: and, lastly, to Mr. Crowther, draughtsman in the Public Works Department, for supplying me with a map of the colony and its Hinterland. The description of the disease will be divided into the following headings: (1) the General Description of the Disease; (2) the Description of Cases of the Disease; (3) the Treatment; (4) the Morbid Anatomy; (5) the Ætiology; and (6) the Geographical Distribution.

THE GENERAL DESCRIPTION OF THE DISEASE.

Henpuye starts in a native of West Africa during or soon after an attack of yaws in which there is a history of the nasal mucous membrane being attacked as a small bony swelling symmetrically placed on either side of the nose. This swelling, which is generally oval with the long axis directed downwards and outwards, is attached to the nasal bones, the nasal process of the superior maxilla, and also to the superior maxilla in the more advanced cases. It is produced by the deposition of new bone under the periosteum on the external aspect of these bones and grows slowly in all directions. It in no way affects the mouth or the orbital or nasal cavities in any case which I have seen, and the nasal ducts are quite unaffected. Rarely the growth is asymmetrical, being situated only on one side of the nose. Pain in the nose and the presence of a sore in that organ are the symptoms complained of at the commencement of the disease; later headache is sometimes felt, and pain in the swelling during wet weather. As the growth becomes larger it seriously interferes with the sight by growing up in front of the eyes and even hiding them, but I have never seen it cause destruction of the eyeball. In many cases the patient has to bend his head downwards in order to be able to see over the tops of the swellings. The skin over the tumor is normal and is freely movable. The course of the disease is that the swellings may cease to grow at any period of their existence or may continue to grow for years—that is to say, they may remain quite small or may grow to be large lumps, in the latter case giving rise to the deformity and the interference with the sight, but I am unacquainted with any case in which they break down or ulcerate. Finally, the disease is much more common, in my experience, in men than in women.

DESCRIPTION OF CASES.

The following cases will be described: (1) slightly developed cases; (2) moderately developed cases; (3) an advanced case; and (4) an asymmetrical case.

Slightly Developed Cases.—CASE I.—The patient, a boy of about seventeen years of age, said that about seven years ago he noticed two small lumps on the nose which began after yaws in which there was a sore in the nose. They increased slightly in size, but soon ceased to grow and have been in their present condition for some years. He never felt any discomfort or pain in them. The two lumps had their long axis directed downwards and outwards, the measurements being half an inch by a quarter of an inch. They were attached to the nasal bones just above the cartilages and the nasal process of the superior maxilla, and were firm, smooth, bony tumors. The skin over them was quite normal and they did not in any way project into the nasal cavity or affect the line of vision, being too small for the latter purpose. There was very little deformity and no treatment was necessary. In this case the lumps soon ceased to grow.

CASE II.—A small Grunshi girl from Kumassi, about seven years of age, who had had yaws some time previously, felt pain in the nose a few months ago and noticed a small swelling on each side of the nose, and this gradually increased in size till it reached its present condition. Her mother was most anxious to have it removed on account of the deformity. On inspection there was found to be an oval swelling on each side of the nose, attached to the nasal bones and the nasal process of the superior maxilla. The long axis of the swelling was directed downwards and outwards—an inch in length and half an inch in breadth. The nasal cartilages were not affected and the interior of the nose was normal. The orbital cavity, the mouth, and the nasal ducts were quite unaffected. The skin over the swelling was normal and freely movable. The patient felt no pain in the tumor and she had never had any headache. The growths were removed by operation. It was very difficult to obtain definite history as to the time when this patient had had yaws and as to the time when the growth appeared, but as far as I could make out the yaws were well developed when the swelling was first noticed.

Moderately Developed Cases.—CASE III.—A young man, a Ga native, who had had yaws about seven years ago, felt pain in the nose and got a person to look into it, who said that there were yaw spots on the mucosa, and later a small swelling on each side of that organ was noticed. These small swellings grew slowly to their present size, and the patient said that they were still increasing. He complained of frontal headache and of slight pain in the swellings in wet weather. On inspection two symmetrically placed swellings were seen on each side of the nose, looking somewhat like small eggs. They were oval in shape, with the long axis directed downwards and outwards. The left measured two inches by two inches and the right three inches by two and a half inches. A profile view showed that they were slightly concave on the side towards the orbit. They did not affect the orbital or nasal cavities, nor did they project into the mouth or affect the nasal ducts or the cartilages of the nose. They were attached to the nasal bones, the nasal process of the superior maxilla, and to the superior maxilla itself. They were smooth, but on the left side the tumor rose to a central ridge. The skin over the swellings was quite normal and was freely movable. In order to see clearly, the patient often had to bend his head somewhat. The growths were removed by operation.

CASE IV.—The patient was an Akwapim woman, aged about twenty years. This case was similar to Case III., but the swellings, which had started when the patient (who had suffered from yaws) was seven years of age, were rather more rounded. She would not consent to operation.

An Advanced Case.—Case V.—A man, a native of Appolonia, about forty years of age, stated that the swellings began with pain in the nose after yaws, when he was about six years old. They grew steadily and slowly till eight years ago, when they stopped, and they have not increased in size since then. On inspection there were two oval swellings situated on each side of the nose, the left measuring two and a half inches by one inch and the right three-quarters of an inch by half an inch. They projected upwards over the orbit, the long axis in each case being directed downwards and outwards. They did not project into the mouth, the nose, or the orbit, and the nasal duct was free. They were attached to the nasal bones, the nasal process of the superior maxilla, and to the maxilla itself. The skin over the tumor was normal and it was freely movable. The patient complained of headache and found that the swellings interfered with his vision considerably, particularly on the left side. He refused to submit to operation.

An Asymmetrical Case.—Case VI.—An Ashanti boy, aged six years, from Donkeo Inquanta, had yaws, and while suffering therefrom, just a year previous to his consulting me, the swelling appeared on one side of the nose, and had been growing ever since. There was no sign of any lump on the other side. He was advised to go to Kumassi for operation.

THE TREATMENT.

I have attempted to reduce these swellings by the administration of iodide of potassium, but have not met with any success. The only treatment appears to be the removal by operation. The method I adopt is as follows. The eyes being protected by a pad over each, an incision is made along the long axis of the tumor and the skin is freed on all sides so that its base is exposed. If the swelling is very small in a child it may be necessary to make a cross cut through the skin as well, in order to get sufficient room to work in. The bone being exposed, a portion of the swelling can easily be cut away by bone forceps, because it is very soft. If large, a few nicks with a Hey’s saw are found most useful in enabling a large portion of the mass to be removed entire. After as much has been removed as possible with the bone forceps, more may be got away by means of the gouge or the gouge forceps or the nibbling forceps. I have experienced difficulty in removing the deeper portions, particularly those close to the orbit. I need hardly say that in the latter the eye has to be carefully guarded from injury. After removal of the bone the wound is well washed out with an antiseptic lotion. The bleeding is slight and is easily controlled by pressure. The wound is closed by a continuous suture and it heals up readily.

THE MORBID ANATOMY.

I have never had any chance of examining the growth *post mortem*, but the portions which I have removed *en masse* by operation have enabled me to make some investigations. The periosteum strips off readily, and under this is a thin shell of compact bone, which appears somewhat ridged on the side towards the periosteum. The rest of the tumor consists of cancellous bone. The whole swelling cuts readily with bone forceps and consists of quite soft bone. On making microscopical preparations there were signs of ossification in membrane proceeding under the periosteum, and the rest appeared like ordinary wide-meshed cancellous bone. The whole process appeared to be that of a slow “osteoplastic periostitis.”

ÆTIOLOGY.

Two views on the ætiology of this disease have been brought forward up to the present time, as far as I know—viz., that the swellings were of a racial character and that the process was started by the larva of some insect. With regard to the first I have only to mention that the disease is found in Ashantis, Grunshis, Fantees, Abantas, the Ga people, etc., races quite different from one another, to show that this cannot be entertained. As to the second, I have never met with evidence which would support the idea that the disease was started by a larva. On the other hand there is always the history of yaws and of the tumor starting during the attack of yaws—*i. e.*, during the period of eruption or soon after. Then, again, the patients complain of pain in the nose with, in some cases, distinct history of a sore and sometimes discharge preceding the swelling. This might be due to some irritation or ulceration of the nasal mucous membrane by the yaws. I have never had the opportunity of examining any person at this stage of the disease, but in the more developed cases I have examined the nose for marks or signs of old ulceration, but have not found them. If, however, the nasal process of the superior maxilla be examined a few foramina are to be seen, and these are often joined together by a small groove indicating the position of a bygone suture. The foramina are for small bloodvessels, which are said to communicate with those of the mucosa of the nose. The site of these foramina is the situation where henpuye starts, and I venture to bring forward the theory that the causation of this peculiar disease is due to an osteoplastic periostitis brought about by the absorption of the poison of yaws from the nasal mucous membrane through the small vessels (or lymphatics) keeping open the foramina which indicate the suture above mentioned.

THE GEOGRAPHICAL DISTRIBUTION.

I am only aware of cases reported from the Gold and Ivory Coasts of West Africa and the West Indies. I never met with it in Mamprusia, nor have I met any trader coming from Moshi with it, nor have I met with it in Fra Fra, and I can find no one who has seen it in the eastern parts of the colony. But in the following districts it has been noted: Ahanta, Appolonia, Fantee, Accra, Aquapim, Akim, Assin, Sefwhi, Ashanti, Attabubu, Kwahu, Kintampo, Berekum, Gaman, the Neutral Zone, and Wassaw. It is perhaps most common in the Sefwhi, Wassaw, and Appolonia districts which adjoin the French Ivory Coast, where cases are also known.

I look upon henpuye as a localized osteoplastic periostitis in the region of the nasal process of the superior maxilla, generally symmetrical, due to yaws, and found among the natives of West Africa and the negroes of the West Indies.

THE MADDOX ROD OR THE PHOROMETER; WHICH?

In the last issue of the Journal there appeared an abstract with the above title, and believing the subject to be of much interest at the present time, our readers have been invited to send us their opinions on the matter, as based on the experience obtained in practice. The communications below have been received and are presented in the order of their reception. We shall be glad to hear from any physicians who are interested [Ed.].

Dear Dr. Deady: In reply to your favor requesting my opinion regarding the respective merits of the Maddox rod and the diplopia test, I wish to say that my experience leads me to rely more and more upon the obscuration test, and while I have not followed out the comparison to any great extent, such as is shown by your tables, results obtained by relying upon the rod test in the detection of heterophoria, as well as in determining when the weak muscles have been sufficiently developed, have been such as to warrant my continuance of its use.

E. D. Brooks.

I have with interest watched the discussions of late, as to the relative value of the Maddox or Stevens tests for heterophoria, as I have for years used them both.

My muscle tests have been made for the last five years at least, with a Risley phorometer, which combines both tests upon one arm and has proven for me a most satisfactory instrument.

I am sorry to say that I have not kept any comparative statistics of my examinations; at the same time they have all left an impression upon my mind, which is this: that I feel more confidence in the results obtained from the use of the Maddox test in the routine tests that I always make of refractive cases. If this test shows any marked degree of heterophoria it has been my habit to retest the patient by the Stevens method, which is usually the same, provided the patient has a sufficient amount of intelligence to give correct answers to the questions put to him. During this test the patient is allowed to sit for some time in front of the prisms, and the eye muscles allowed to relax from that first impulse at muscular effort that follows the placing of the prisms in front of the eyes.

To my mind both tests are good and fairly accurate in the hands of one who is thoroughly familiar with their use and shortcomings, provided your patient is able to answer correctly.

Many times, on re-examining a patient, I have discovered what appeared to be a great change in the muscular conditions, but after repeated examinations I have usually found it was the patient, and not the muscles, that was erratic.

When Dr. Hubbell speaks of ¼° of difference between the Maddox and Stevens tests, he has more confidence than I have in the average judgment of patients that come under our care.

SAYER HASBROUCK.

DEAR DOCTOR: Your note asking my opinion of the comparative usefulness of the Maddox rod and the phorometer is at hand.

In the detection of heterophoria I regard the rod as the most convenient and trustworthy instrument used.

The distance at which the test is made and the dissimilarity of the images seen usually eliminate all actual effort to hold the eyes in any particular position other than that in which they stand the most easily. Accordingly the deviation is quickly noted and readily measured.

So satisfactory has this modest little instrument been in my examinations that I now rarely resort to other methods. The amount of deviation sometimes shown between this and other instruments is so slight as to make little or no difference in the measures employed for correction.

It is to be noted that cases not unfrequently occur in which a hyper-sensitive, or, on the contrary, an enervated condition exists, which is not fully indicated by any instrument. An educated judgment will here have to supply conclusions not to be drawn by any hard-and-fast rules.

After the rod and the phorometer came into use and an opportunity was presented to compare the results obtained by each, I made a careful test of eighty pronounced cases of errors of refraction accompanied by heterophoria. Of this number only nine showed a persistent difference of deviation and in none of them a difference greater than 1½°. But this was not always on the one side or the other, as six out of the nine showed a higher degree of deviation by the rod than by the phorometer. Eighty cases may not be enough upon which to base an orthodox conclusion; but my experience with the rod has been so satisfactory that I now seldom use the phorometer at all. It appears quite possible practically to estimate the degree of heterophoria as accurately with the one instrument as with the other; and while it is true that a correction of the error of refraction will commonly correct the deviation, still all cases of optical defect should be tested with the rod or phorometer before the lenses are prescribed.

WM. A. PHILLIPS.

MY DEAR DR. DEADY: Dr. Hubbell limits the discussion "to the comparative value of the diplopia test, by Stevens' phorometer" and the Maddox rod test.

It would be interesting to follow out the idea with other phorometers,—and with the Wilson phorometer my records do not show quite such a marked difference in results,—but I have not taken pains to get comparative results in any considerable number of cases.

Dr. Hubbell says: "In the diplopia test, the dissociation is effected by changing the visual axis of one eye by means of a prism. The displacement of one image cannot be done without associating with it, more or less, an impulse to some form of ocular effort.... In the obscuration test (Maddox rod) no such effort is invited, no change of innervation takes place." But in the rod test the light seems nearer to the patient than in the prism test. This may account for much of the difference in results and amount to "an extraneous impulse to muscular contraction."

Dr. Hubbell is entirely justified in his conclusion as made upon experiments with the Maddox rod and the Stevens phorometer. I shall watch cases along similar lines with the Wilson phorometer and report later.

In the mean time the rod and the prism tests may well be taken in each case and let judgment decide as to treatment.

THOS. M. STEWART.

I agree with the writer that the rod test is the more scientific test for heterophoria, and of late years have virtually discarded the prism test, except in special cases. The tables are interesting, but their value would be materially increased if the author would supplement them with tables showing the refraction, and inflammation or its results.

Was it an accident that Stevens' phorometer showed the same amount of right hyperphoria in one-ninth of the cases, and in thirteen of thirty-three cases of left hyperphoria? In which of these cases was there anisometropia and of what kind was it?

What was the refraction of the two cases of exophoria, two of left and one of right hyperphoria by the phorometer; and was the refraction the same in the six cases which were orthophoric by both rod and prism?

Such studies are necessary to a clear understanding of the relative value of these tests.

JOHN L. MOFFAT.

DEAR DR. DEADY: Your letter and inclosed article on "The Maddox Rod or Phorometer; Which?" has been received and examined with interest.

I have examined a good many cases in my office by both methods and find variable results, but where there is a radical difference I have found the Maddox rod the more accurate, and from experience I have learned to rely upon it instead of the phorometer, as in prescribing prisms in hyperphoria in connection with glasses for constant use I rely wholly upon the rod test.

J. M. FAWCETT.

DEAR DOCTOR: Concerning the discussion of Maddox Rod vs. Phorometer about which you wrote me—can say that I believe that the Maddox rod is the more reliable test. My reasons on theoretical grounds for so believing are briefly these.

Given a case for examination; the test which *least disturbs* the muscular co-ordination under investigation must give the best result. Now I think that when we throw the images into non-corresponding retinal points that we almost certainly cause some tension of certain muscles, because it is putting the eyes in an *unnatural* relation with one another; and this is done by the phorometer. The Maddox rod is theoretically free from this objection.

Practically the deviations are more certainly measured, because a patient *knows* when the streak cuts the light; and you cannot trust their eye alone to tell when the lights are exactly in a line. Have used *both* tests in every case I have examined in my private practice, and I find the Maddox the more reliable test. It is more to be depended upon.

EDW. HILL BALDWIN.

ABSTRACTS FROM CURRENT LITERATURE.

Grant, Dundas.—Case of Emphysema of the Orbital Wall of the Anterior Ethmoidal Cells, Caused by blowing the Nose.—*Jour. Lar., Rhin. and Otol.*, March, 1900.

This case was shown to the British Laryngological, Rhinological and Otological Association.

W. M., twenty-eight years, came under my care yesterday on account of a sudden swelling of his eye which had taken place two hours previously, and which had occurred suddenly as he was blowing his nose without a handkerchief, and which gave him the impression as if something were running out of his eye. The swelling crackled in a manner characteristic of emphysema, and the first suspicion was that he must have had some disease of the orbital wall of the anterior ethmoidal cells, and that on examination there would be found some evidence of ethmoidal disease. None such was to be elicited, and the only history obtainable was that he received several kicks on the nose and back of the ear two months ago. This has probably resulted in a fracture of the orbital wall of certain of these cells.

PALMER.

Lack, Lambert.—Case of Nasal Polypi, with Suppuration and Absence of Maxillary Sinuses.—*Jour. of Lar., Rhin. and Otol.*, April, 1900.

A man, aet. twenty-eight years, complains of nasal obstruction and purulent discharge, with a disagreeable odor in the nose. The polypi having been removed, the pus appeared to flow from under the anterior ends of the middle turbinates. After wiping the discharge away and bending the patient's head forward, it reappeared in large quantity. On transillumination the cheek on both sides appeared quite dark, and the patient had no subjective sensation of light. The diagnosis of antral suppuration was now considered almost certain, and the patient was advised to have both antra punctured from the alveolar margins. This was accordingly attempted under gas, but although the antrum drill was forced in for its full length, no cavity was reached.

Puncture from the inferior meatus was next attempted, and considerable force was used in two different points; but with no better result. It would seem therefore that the antra must be very small, if not entirely absent.

Discussion.—Mr. Spencer thought it might be one of those convoluted inferior turbinals which form a gutter in which pus collects. The majority considered it suppuration in the ethmoidal region.

PALMER.

Lawson, Arnold.—Cicatrix Horn Growing from the Cornea.—*The Lancet*, February 3, 1900.

The patient was a female child, aged eight years, a hydrocephalic idiot. The history given was that about one year previously a white spot had appeared on the right eye and that the eye began to project. Six months later a growth was first noticed on the right cornea, and this had constantly increased in size. Latterly a white spot had appeared on the left eye. On examination of the eyes there was seen a large conical tuberculated excrescence protruding between the lids of the right eye. It was half an inch in length and its base attached to the cornea covered about four-fifths of its surface. The left cornea exhibited a yellowish infiltration just below the pupil, over which the cornea was bulging; the anterior chamber was deep, the iris was immobile, the tension was slightly raised, and the eye was quite blind. Both globes were very anæsthetic, and there was considerable muco-purulent discharge from a chronic inflammation of both conjunctival sacs. The growth upon the right eye was accidentally detached a few days after admission into the hospital, and it was then seen to have been attached to the cornea at the apex of a central staphyloma, which was left covered by a fleshy soft core which had formerly been embodied in the center of the growth. The cornea was entirely opaque, and the eye was quite blind. After removal of the right eye a few days later examination of the globe revealed a co-arct retina with evidences of chronic degenerative changes in all the various structures. The anterior chamber was completely abolished, the iris throughout its extent being firmly adherent to the back of the cornea, which was bulging centrally. The apex of the corneal staphyloma had evidently been the site of a large perforation, which was closed by the fleshy granulations which formed the core of the growth. The growth itself measured half an inch from apex to base and one and a half inch around its base.

The interior portion was soft and crumbling, but the external layers were hard and horny and cut with difficulty. A wedge-shaped piece was cut away from the growth and specimens were cut and stained with carmine. The microscope showed that the external layers consisted of several faintly fibrillated strata of a dense, homogeneous nature. The layers occupied about one-quarter of the entire thickness of the walls, the rest being entirely composed of small nucleated cells, those most external being stratified. Adopting Mr. Bland Sutton’s classification of human horns, this growth would be an example of a cicatrix horn, the rarest of all varieties of horn, and one which had been usually found in connection with cicatrices of burns and scalds. The probable ætiology in this case was an overgrowth of granulation tissue closing the perforation in the cornea, which, owing to an unhealthy condition of the wound and eye, which was anæsthetic and atrophic, had become exuberant, simulating exactly the condition known as “proud flesh” elsewhere. By a process of accumulation and heaping up, the granulations gradually formed a cap over the cornea, whilst the external layers gradually became stratified and horny from the pressure of fresh growth from the central core and by the action of the air. The nature of the growth was evidence that the corneal epithelium bore no share in its production and discounted the possibility that it might be due to a huge crust of inspissated conjunctival discharges.

DEADY.

Lodge, Jr., M. D., Samuel.—A Case of Fatal Sphenoidal Suppuration.—*The Laryngoscope*, March, 1900.

W. S., aet. thirty-one years, admitted to Royal Halifax Infirmary May 15, 1899, complaining of pain in right ear and right side of face of six months' duration. For two months right side face swollen and copious bloody, purulent discharge from right nostril. Nine years ago had syphilis. Insomnia from pain.

On admission: Temperature 100°; skin over right superior maxilla red and œdematous; thick purulent discharge from right superior meatus, sequestrum in region of right cribriform plate; naso-pharynx, chest, and abdomen normal; urine, sp. gr., 1014; trace of albumen. Fundi (of eye) normal.

May 16—No pus found in antrum on exploration and flushing. Patient taking 60 grs. pot. iod. (t. i. d.) and mercurial inunction. Temperature in ear usually higher than that in mouth until just before death. June 8.—Mortuus est.

Post-mortem Examination.—Skull. Base of brain was bathed in thick greenish pus, principally in the neighborhood of the pituitary body, the pus extended back over the pons and medulla. No brain abscess. Ventricles contained more than normal quantity of fluid. Frontal sinuses and cribriform plate of ethmoid and ethmoidal cells normal.

To right of the sella turcica there was some necrosis of the walls of the sphenoidal sinus. Probe readily passed from base of skull through sphenoidal sinus into the nose. Large free opening from said sinus into nose, which sinus was full of muco-pus. Cavernous sinus not thrombosed. Right antrum of Highmore contained about a dram of thick glairy mucus.

PALMER.

Killian, Prof. Gustav.—Case of Acute Perichondritis and Periostitis of the Nasal Septum of Dental Origin.—*Münch. med. Wochen.*, No. 5, 1900.

There have been recorded two cases of perichondritis of the septum due to alveolar periostitis. Suppuration of dental cyst was cause in the following case.

A young man had pain in second left upper incisor; two days after obstruction of nose supervened, with pain in forehead and high fever. There was a sudden copious discharge of fetid pus from right nostril seven days later. The entire mucosa of the septum was raised from the cartilages, etc. It is considerably swollen over right side of the triangular cartilage, but less so posteriorly. Severe headache in forehead and frontal eminence, and still little fever. The pus was escaping through a small hole into the left nostril. It was freely incised. The triangular cartilage was disintegrated, and the pus had burrowed between the soft tissues and vomer and vertical plate of the ethmoid. The choanæ were constricted by thickening of the septal mucous membrane. The wound healed in a fortnight without sequestrum, while the toothache lasted but two days.

Six months later the patient had recurrence of pain in the same tooth of two months' duration; it was extracted and pus continued to exude from the socket. A probe, passed 2½ centimeters to the floor of the nose and septum, showed a cavity covered with membrane in the anterior parts of upper jaw, which was a cyst at the root of the tooth. The anterior cyst walls were removed with bone forceps, and the remainder scraped. The cavity gradually healed.

The cyst probably broke through under the septal mucous membrane. In exceedingly few cases of perichondritis does the process extend to the osseous septum. Only once has the author seen record of a case which was as extensive as this. The offensive odor also points to a dental origin.

PALMER.

Hawthorne, C. O.—The Eye Symptoms for Locomotor Ataxia, with a Clinical Record of Thirty Cases.—*Brit. Med. Jour.*, March 3, 1900.

It is now generally recognized that the disease known as locomotor ataxia may include among its clinical manifestations symptoms other than those which depend on pathological changes in the spinal cord. A number of these are associated with the functions of the eyeballs. The Argyll-Robertson pupil is universally admitted as valuable confirmatory evidence of a diagnosis of locomotor ataxia; ocular paralyses, if less frequent, are certainly not less significant; and optic nerve atrophy is at least so well known in connection with the disease that its occurrence in any individual case would hardly call for comment.

A further step forward in our knowledge of the clinical possibilities of locomotor ataxia has been the recognition of the fact that ocular disturbances may precede the evidences of any spinal lesion. This advance necessarily means that the occurrence of any one of the ocular events above mentioned must, unless otherwise explained, generate the suspicion that the case may in its later events display the phenomena known to depend upon sclerosis of the posterior columns of the spinal cord.

It is very difficult to collect the evidence necessary to show in what proportion of cases this suspicion is justified by the event. For it is certain that ocular disturbances may long precede the manifestation of spinal symptoms. In the case of optic atrophy the interval may, according to Gowers, extend even to twenty years. Thus it can only be in very exceptional instances that one and the same physician will have the opportunity of observing at least a number of these cases through all the stages of their progress. Yet, if true, it is of manifest importance, for the sake both of exact knowledge and of accurate prognosis, that it should be clearly recognized that an optic-nerve atrophy, an ocular paralysis, or a loss of the pupil light reflex, unless capable of other explanation, belongs in all probability to the order of events incident to locomotor ataxia, and that any one of these may well be the introduction to a more widely-spread manifestation of the disease.

For reasons stated above, the collection of complete histories necessary to afford actual demonstration of the truth of these propositions is difficult; and all the more so as there is reason to believe that in those cases in which the early stress of the disease falls upon the nervous apparatus of the eyeball the spinal symptoms are apt to be slight in degree as well as delayed in development. This is certainly the case when the ocular disturbance takes the form of optic-nerve atrophy. "In a large number of such cases," says Gowers, "ataxy never comes on, the spinal malady becoming stationary when the nerve suffers."

Of course, in a given case of optic-nerve atrophy without spinal symptoms the question may fairly be raised whether it is right to place such a case in the locomotor ataxia group. All that can be said in reply is (1) that from cases of optic atrophy pure and simple one passes by an unbroken series of steps through cases with more and more distinct evidence of locomotor ataxia to, at the end of the series, optic atrophy in association with characteristic ataxic symptoms, and (2) that, as already stated, a simple case of optic atrophy may remain unchanged for many years, and yet in the end display undoubted evidence of the development of a spinal lesion. But if optic-nerve atrophy may be the primary symptom in the disease, if the occurrence of spinal symptoms may follow it after an interval of many years, and if again it may remain without at any time any existing ataxia, it is not unreasonable to presume that both the Argyll-Robertson pupil and an ocular paralysis may each have exactly corresponding relations to the development of the spinal evidences of locomotor ataxia. The collection of evidence to support this suggestion is even more difficult than in the case of optic-nerve atrophy. The latter condition must ere long compel the patient to seek medical advice, and thus the opportunity for a complete investigation of the state of his nervous apparatus is afforded at a relatively early date. But an Argyll-Robertson pupil may exist, and presumably exist for years, without any inconvenience to the patient. Such a patient, therefore, will not consult his medical adviser until spinal or other symptoms display themselves, and thus the precedence of the pupillary condition cannot be determined. In the case of an ocular paralysis medical assistance is, no doubt, usually promptly invoked. But such an occurrence is open to a number of ætiological explanations, for example, rheumatism, cold, etc., which it is difficult to exclude with confidence. Hence it is much less precise in its significance than either a double optic atrophy or the Argyll-

Robertson pupil. It must be by the collection of observations extending over a long term of years that actual demonstration of the relationship of the ocular disturbances now in question to the occurrence of spinal disease can be established. But while falling short of the merit of actual demonstration, the presentation of the facts displayed by a number of cases which could only be observed over relatively brief periods is not without value. If no one case affords a complete history of all the stages of the disease the picture presented may none the less be fairly complete, provided the cases are sufficiently numerous, and they are seen at different points of development. It is believed that in the present series these conditions are fulfilled. The conclusions they afford, as far as the present purpose is concerned, are: (1) That an optic-nerve atrophy, an ocular paralysis, or an Argyll-Robertson pupil may exist as an isolated symptom for a considerable time, presumably for years; (2) that any two of these may be associated together, with a correspondingly increased presumption that the diseased process causing them is of the locomotor ataxia order; (3) that any one of the three, or a combination of two or all of them, may exist in conjunction with a greater or less degree of evidence of spinal disease; and (4) that occasionally a case which commences with purely ocular symptoms may be seen to develop with comparative rapidity characteristic symptoms of the spinal lesion of locomotor ataxia. The cases therefore may be held to justify the view that an optic-nerve atrophy, an ocular paralysis, or the Argyll-Robertson pupil (not capable of other explanation) must be regarded as affording a definite basis for suspicion in reference to a possible development of spinal disease. On the other hand, it must be admitted that the prognostic indication, so far as spinal disease is concerned, is not an absolute one, for the ocular defect may exist certainly for many years without any evidence whatever of the involvement of the spinal cord.

The cases here recorded have all been the subject of detailed and in most cases repeated examination, and unless the contrary is stated, it may be taken for granted that the thoracic and abdominal viscera are normal, to physical and other methods of examination. In all cases, too, in which no specific statement is made, it is to be understood that the visual acuity, the visual fields (both for white and colors), and the fundus oculi have been proved to be normal. This last statement of course does not apply to cases in which optic atrophy exists. Particular care has been taken to be accurate in regard to the condition of the pupils and the knee-jerks. In nearly, if not absolutely in every instance where a departure from the normal is chronicled, the record has been confirmed by more than one observer, and in the case of a deficient knee-jerk the conclusion stated has never been formulated until the conditions insisted on by Gowers, Buzzard, and Jendrassik have been fulfilled. With a few exceptions in which only a single observation was possible, the patients have been watched for months, and in some instances for several years. The cases are arranged in series, with a view to show how, from a purely ocular condition, one may pass through gradually accumulating evidence to the same ocular condition in association with the characteristic signs of the spinal lesion of locomotor ataxia.

I.—Cases in which Optic-Nerve Atrophy is the Primary or Dominating Condition.

(*a*) *Optic Atrophy, without Other Evidence of Disease.*

Case I.—W. T., aged twenty-five. Failure of vision extending over two years, with reduction of visual acuity to the power of counting fingers at three feet. Double optic atrophy; pupils medium, with distinct light response; knee-jerks distinct and no evidence of spinal disease, and no cerebral symptoms other than one or two attacks of giddiness. Urethritis, but no syphilis.

Case II.—F. R., aged thirty-eight. Double optic atrophy, with almost complete loss of vision, the defective sight having been observed for at least eighteen months; pupils dilated and immobile; no evidence of spinal disease, unless possibly some degree of failure of sexual power; no cerebral incidents; no history or evidence of syphilis.

(*b*) *Optic Atrophy, with Other Ocular Evidence Suggestive of Locomotor Ataxy.*

Case III.—(By permission of Mr. Ernest Clarke, F. R. C. S.) R. C., aged thirty-nine. Double optic atrophy, reducing right visual acuity to the power to count fingers at four feet, and left to mere perception of light; right pupil dilated and three times the size of the left; neither any light response, but free movement on convergence; entire absence of symptoms and objective signs of spinal disease; “gleet” twenty years before, no syphilis.

CASE IV.—A. S., aged twenty-five. Double optic atrophy, with observed failure of vision for twelve months. V. A. right-hand movements only; left, $\frac{6}{18}$ part; pupils 2.5 mm., no light response, but contract on convergence; knee-jerks difficult to obtain, but movement, though possibly wanting in promptness, is normal in extent; no ataxia or other evidence of spinal disease; mother of three healthy children, no miscarriages.

(*c*) *Optic Atrophy, with Some Evidence of Spinal Disease.*

CASE V.—W. A., aged thirty-seven. Double optic atrophy, with reduction of visual acuteness to “hand movements;” pupils dilated and immobile; knee-jerks absent, but no other evidence of spinal disease; venereal sore when aged twenty; no recognized secondaries, and father of four healthy children.

CASE VI.—J. G., aged thirty-five. Failure of sight (six months); optic atrophy, gradually increasing whilst under observation of twelve months; pupils not definitely abnormal; knee-jerks absent throughout, but no further appearance of spinal disturbance; urethritis, but no history of syphilis; father of two healthy children, wife no miscarriages.

CASE VII.—F. L., aged thirty-nine. Double optic atrophy, reducing visual acuteness to $\frac{6}{24}$, pupils very small, and with Argyll-Robertson phenomenon; subsequent to failure of sight (twelve months) has had shooting pains in thighs, and failure in retention power of bladder; knee-jerks distinct; no ataxia or sensory defect in lower limbs; venereal sore twenty years before; no recognized secondary syphilis; wife healthy: seven pregnancies, five miscarriages.

(*d*) *Optic Atrophy, with Distinctive Evidence of Spinal Disease.*

CASE VIII.—C. H., aged thirty-eight. Failure of sight (two years) from double optic atrophy; pupils medium, with Argyll-Robertson phenomenon; moderate double ptosis, but no ocular paralysis; shooting pains in lower limbs (eight years); knee-jerks absent; considerable ataxia and failure of control over bladder; syphilis at nineteen years.

CASE IX.—G. S., aged forty. Pallor of disks and peripheral contraction of visual fields; four months later loss of knee-jerks and gradual development of ataxia; pupils normal throughout; death at the end of twelve months with symptoms of meningitis; syphilis at twenty-five years.

II.—CASES WITH ARGYLL-ROBERTSON PHENOMENON.

(*a*) *Argyll-Robertson Phenomenon, without Other Evidence of Disease.*

CASE X.—A. L., aged thirty-three, the subject of slight hypermetropic astigmatism. Pupils small, not quite circular, with Argyll-Robertson phenomenon; no other ocular defect, and no evidence of a spinal lesion. No history of syphilis.

CASE XI.—K. S., aged forty-three. Pupils rather small, unequal, quite destitute of light response, though moving freely in convergence; no other ocular defect except some presbyopia; no evidence of spinal disease, though left knee-jerk not easily obtained. Unmarried; syphilis seems highly improbable.

CASE XII.—G. G., aged sixty. Pupils small, with distinct Argyll-Robertson phenomenon. Knee-jerks, not easily obtained, but not definitely abnormal, and no other evidence of spinal disease. Patient suffers from defective vision, probably from tobacco poisoning (central scotoma for red); no history or evidence of syphilis.

(*b*) *Argyll-Robertson Phenomenon, with Other Ocular Disturbance Suggestive of Locomotor Ataxia.*

CASE XIII.—D. T., aged forty-two. Pupils below medium size, destitute of light response, with free movement in convergence; had for seven days suffered from diplopia, and under observation gradual development of complete paralysis of right external rectus; no other ocular defect. Knee-jerks distinct, and no suggestion of spinal disease; chancre of lip and secondary syphilis nine years before.

(*c*) *Argyll-Robertson Pupils, with More or Less Evidence of Spinal Disease.*

CASE XIV.—T. F., aged fifty-five. Right pupil 2mm., left 3 mm., each with Argyll-Robertson phenomenon; no other ocular defect except presbyopia. Ten years ago had difficulty in passing urine, and since then occasionally voids it involuntarily, and for eighteen years has been liable to seizures of pain in calves, insteps, and heels; knee-jerks normal, and no objective signs of spinal disease. Venereal sore when aged twenty, but no secondary symptoms, and father of six healthy children.

CASE XV.—H. W., aged thirty-eight, is the subject of hypermetropia, 4.5 D. Pupils very small, especially left; neither moves under light, but distinct contraction during convergence. Admits recent difficulty in descending stairs, saying he "frequently misses the bottom step," and has suffered from "sciatica" for two years. No objective evidence of spinal disease, and urinary and sexual functions undisturbed. Admits gonorrhea, but denies syphilis. Wife miscarried eight months after marriage; no further pregnancies.

CASE XVI.—R. S., aged forty-three. Hypermetropic and presbyopic. Pupils small, unequal, not quite circular, and with definite Argyll-Robertson phenomenon. Knee-jerks cannot be obtained (confirmed on three different dates), but no other sign or symptom of tabes dorsalis, unless "rheumatic pains" in lower limbs for several years. Unmarried; no history of syphilis. Four years ago had, after "catching cold," to have urine withdrawn by a catheter, but no subsequent disturbance of bladder function.

CASE XVII.—E. W., aged forty-eight. Myosis with Argyll-Robertson phenomenon; right ptosis and crossed diplopia (one month), without obvious ocular paresis; absence of right knee-jerk (confirmed on two occasions), and failure in retention power of bladder (six months), but no other evidence of spinal disease; vulvar sores and skin eruption six years before.

III.—CASES IN WHICH AN OCULAR PARALYSIS IS THE EARLIEST OR DOMINATING SYMPTOM.

(a) *Ocular Paralysis, without Other Evidence of Disease.*

CASE XVIII.—(By permission of Mr. J. T. James, F. R. C. S.) F. D., aged thirty-seven. Dilated and immobile pupils, without any other ocular defect.

No evidence of spinal disease; syphilis nine years ago; no change while under observation for three years.

CASE XIX.—K. K., aged twenty-eight. Iridoplegia, double, followed by paralysis of left external rectus, the condition being under observation for nearly a year, but without the discovery of any satisfactory explanation. No evidence of spinal disease. Married, four healthy children, no miscarriage; during one pregnancy very free loss of hair (now grown again), but no other occurrence to suggest syphilis. No family or personal history of gout or rheumatism.

CASE XX.—G. H., aged thirty-three. Dilated and immobile pupils, with incomplete ptosis and divergence of eyeball on each side. Present condition of four years' duration, and separated by an interval of three years from venereal sore and skin eruption. No other ocular defect; no evidence of spinal disease, and general health good throughout. No change while under observation for three months.

(b) ***Ocular Paralysis, with Other Ocular Evidence Suggestive of Locomotor Ataxia.***

CASE XXI.—E. S., aged forty-three. Diplopia from paralysis right external rectus, pupils small, each with Argyll-Robertson phenomenon; visual left acuity only $\frac{6}{12}$, and small but distinct central scotoma, with some contraction of the peripheral field; knee-jerks distinct, and no ataxia or other evidence of spinal disease. Three early miscarriages, no full-time child. No change while under observation for nine months, but on two occasions severe attack of vomiting and abdominal pain, extending over several days and without recognized cause (? gastric crises).

CASE XXII.—A. M., aged fifty-five. Left ptosis and paralysis of external ocular muscles supplied by third nerve in 1887, the pupils being normal, followed by incomplete recovery. In 1897 development of identical condition on the right side, and pupils found to be small and with Argyll-Robertson phenomenon; knee-jerks very slight and with great difficulty, but no other evidence of spinal disease. No history of syphilis.

CASE XXIII.—H. F., aged thirty-seven. Right ptosis with diplopia (seven days) and defective inward excursion of right eyeball; pupils very small, not quite equal, and with Argyll-Robertson phenomenon; optic disks pale and marked contraction of visual fields, but normal central vision; knee-jerks scarcely to be obtained, but no other evidence of spinal disease. No history of syphilis. Father of four healthy children. Seen after a month’s interval, paralysis of all external right ocular muscles supplied by third nerve, and knee-jerks absent.

(c) *Ocular Paralysis, with More or Less Evidence of Spinal Disease.*

Case XXIV.—G. S., aged forty-one. Ptosis and complete ophthalmoplegia externa on left side, with dilated and immobile pupils and some degree of right ptosis, these conditions or some of them having been present for five years. Knee-jerks distinct, and no ataxic phenomenon, but imperfect control over bladder, and failure of sexual power during last six months. No admitted syphilis.

Case XXV.—J. L., aged forty-two. Diplopia and drooping left upper eyelid for four years. Ptosis left side, and marked defect of ocular movements in each eye; left pupil dilated and immobile; right small, contracts during convergence, but no light response; no other ocular defect. Knee-jerk scarcely obtained on either side; no ataxia, but attacks of “twitching pains” in lower limbs, and for some time difficulty in starting the flow of urine. Venereal sore in 1882, and subsequent loss of hair, but no other secondary symptoms. Patient watched for twelve months without appreciable change.

Case XXVI.—L. D., aged forty-four. Crossed diplopia (one month), without obvious ocular paralysis, and pupils small with Argyll-Robertson phenomenon. Knee-jerks absent; no ataxia to usual test, but has noticed tendency to stagger in the dark; is troubled with pains in the knees, has difficulty in commencing the act of micturition, and recent marked failure of sexual power; venereal sore at twenty years, and subsequent sore throat, but no skin eruption or loss of hair. Father of three healthy children.

Case XXVII.—J. H., aged forty-seven. Double vision of two months’ duration; similar attack three years ago, with complete recovery. Paralysis of right external rectus; pupils, visual acuity, and visual fields normal. Knee-jerks absent, but no other evidence of spinal disease. Patient the subject of albuminuria, and presents physical evidence of an aneurism of the ascending aortic arch. Youngest child has marked evidence of inherited syphilis.

CASE XXVIII.—W. M., aged thirty-five. Paralysis of left third nerve, without iridoplegia or cycloplegia; pupils normal. Knee-jerks absent, but no ataxia or other evidence of spinal disease. Several venereal sores ten years ago, but no recognized secondary syphilis. There is, however, evidence of a former iritis.

CASE XXIX.—M. C., aged fifty-four. Paralysis of left external rectus, with history of two previous attacks of diplopia during last four years; no other ocular defect, unless some imperfect light response in left pupil; knee-jerks absent, and complaint of "sciatica" for two years, but no other evidence of spinal disease. Albuminuria distinct, and physical signs of hypertrophy of left ventricle. No history of syphilis.

CASE XXX.—(By permission of Mr. N. M. MacLehose, M. B.) H. Y., aged thirty. Homonymous diplopia observed over a period of six months without appreciable ocular paralysis; pupils of medium size, with definite Argyll-Robertson phenomenon; knee-jerks absent, and in later months decided ataxia and sensory defects in lower limbs; visual acuity unaffected to ordinary test, but gradual contraction of visual fields, especially on right side; chancre and secondary syphilis four years before.

There are in these series of cases many facts which might reasonably be made the subject of remarks, and several of the cases are certainly of great individual interest. But they are here displayed in the above grouping for the purpose of illustrating the clinical order and sequence in which, as a matter of actual experience, the ocular disturbances of locomotor ataxia may manifest themselves in relation to the spinal evidences of that disease. Of course, in those cases in which there exists only a single ocular symptom unaccompanied by any sign of spinal disease, it may be objected that it has yet to be demonstrated that such cases are of the nature of locomotor ataxia. It is doubtless to be desired that such cases should be under exact observation as long as the opportunity for further developments exists—that is, for the entire life of the patient. But to insist upon such a condition is a mere counsel of perfection. One must make reasonable use of such evidence as the brevity of life and the exigencies of practice permit. And the evidence here set forth affords at least a very strong presumption, to say the least of it, of the truth of the doctrines stated in the earlier paragraphs of this paper. Probably the particular proposition which is most likely to be contested is the one which places the Argyll-Robertson pupil equally with optic-nerve atrophy, and an ocular paralysis, as a possible first event in the eruption of the phenomena of locomotor ataxia. But on turning to the records it will be found that the facts support this suggestion almost as strongly as they support the corresponding suggestion in reference to optic-nerve atrophy and ocular paralysis. Attention in this respect may be particularly given to Case XIII. The man complains of a quite recent diplopia, and he has undoubtedly had syphilis; the pupils show the Argyll-Robertson phenomenon. It is in the highest degree probable that, had the patient been under observation a week or two earlier, the condition of the pupils would have been the sole existing ocular abnormality. Yet in the light of the development of an ocular paralysis, it can scarcely be doubted that, whether he develop spinal symptoms or not, his nervous system is the site of diseased processes of the locomotor ataxia order. When to these facts there are added, as in Cases XIV. to XVII., illustrations of the various forms and degrees of evidence of spinal disease that may be associated with the Argyll-Robertson pupil, it seems impossible to resist the conclusions that the condition of the pupil so named may be the first evidence of locomotor ataxia; that it may precede by varying intervals other evidences of the disease; and that at least very probably, in a certain number of cases, the symptomatology of the disease may be permanently restricted to this one event. In some examples of its spinal form locomotor ataxia is undoubtedly an extremely chronic disease, with few and imperfectly developed symptoms; and it is thus not unnatural to expect that similar limitations may obtain in the ocular manifestations of the disease. That evidences of grave nervous disease may be limited to the pupil is well seen in Case XVIII., where a syphilitic patient was under observation for three years without the discovery of any abnormality other than paresis of each sphincter iridis. There is certainly no obvious reason why a similar restriction should not determine the Argyll-Robertson pupil as a purely isolated phenomenon with, it must be added, the same unfortunate possibilities that are undoubtedly attached to the patient whose case has just been quoted. The conclusions above adopted in reference to the Argyll-Robertson pupil are applicable, *mutatis mutandis*, to optic-nerve atrophy and to ocular paralysis, as is abundantly demonstrated in the corresponding series of the cases recorded in this paper.

DEADY.

Menzies, J. Acworth.—Detachment of Corneal Epithelium (?). —*British Med. Jour.*, March 17, 1900.

The following case seems to be worthy of record because of the long duration of the symptoms and the immediate relief ultimately obtained. Mrs. W. consulted me on August 4, 1899, and gave the following history: Five years previously the right eye was struck and “cut” by a cricket ball. Since that time there had been pain exactly as if there was a foreign body under the lid or embedded in the cornea. There was a pricking feeling on winking, and the patient could not bear to have the upper lid touched in its outer half. She could only obtain ease by keeping the eyes closed and perfectly still, or wide open with the lids motionless. On examination no foreign body could be seen, and the lids were normal. In the lower outer quadrant of the cornea careful observation showed that the epithelium was ruffled and freely movable over a small area, and in part of the same area was a tiny circular, slightly opaque, raised patch of the corneal tissue. Nothing more could be made out. I prescribed a bandage and some boric lotion with cocaine. Two months later, on October 6, I again saw the patient, who was then in precisely the same condition as before, and had been so during the two months’ interval. She was in such misery that I decided to adopt surgical measures at once. Accordingly, after instilling cocaine, I carefully explored the painful area with a needle, but could detect no foreign body. I then scraped the part thoroughly with a sharp spoon, removing the epithelium for some little distance around, and a fair amount of corneal tissue in the affected area. The following day there was some smarting, but the eye could be moved freely under the lid, and there was no pain on pressure over the previously tender spot. Progress was uninterrupted. The epithelium grew over the denuded surface, and no opacity resulted. The eye now is perfectly right and the vision is normal.

I should have put the difficulty I had in making a diagnosis down to my having overlooked some detail, had it not been that the patient was for a considerable time under treatment at an eye hospital. The explanation I am inclined to adopt, for want of a better, is this, that the original blow caused the anterior elastic lamina with the epithelium to become detached. The nutrition of the epithelium might thus be kept up, and every movement which pressed upon the surface would bring the detached membrane down on the corneal nerve filaments. But it must be confessed it is not easy to understand how this condition could remain stationary for five years.

DEADY.

Hines, M. D., Oliver S.—Iodide of Stannum in Tuberculosis.—*The Amer. Hom.*, March 15, 1900.

The author thinks iodide of stannum often preferable to stannum in tuberculosis. He uses it when the patient has a clear complexion and long eyelashes and where the progress of the disease is rapid. He reports a case for which the 2x trituration was given, in which there was “a marked tubercular affection of the chest, increased vocal fremitus, an abundance of thick yellow and sweetish sputum, sweat at night, and rapid emaciation.” The result was encouraging.

PALMER.

Kyle, M. D., D. Braddon.—Initial Forms of Tubercular Laryngitis.—*Inter. Med. Mag.*, March, 1900.

The enumeration and exact description of these prodromal symptoms are so important that we copy them in full.

The following, which is a translation of an article by Monsarrat of Paris (*Rev. Hebdom. Laryngol., d'Otol., et de Rhinol.*, No. 43, October 28, 1899), covers the ground so thoroughly that it is worthy of repetition:

"Laryngeal phthisis completely developed presents multiple and varied symptoms, some more characteristic than others. In one patient are found symptoms functionally grave, out of proportion to the lesions relatively benign. In another, physical signs take first place; there may be an ulceration completely obliterating one cord, or considerable œdema of the arytenoids and vestibule, which closes the opening of the glottis. Having reached the period when tuberculosis is easily recognized, the various patients are able to date their laryngitis from diverse pathologic beginnings. This one will present solely the history of a cough, the other a raucous voice, in another pain will take precedence. In mentioning these various modes of commencement we insist on the connection which may exist between each of them and the localization at the beginning of tuberculosis, on one or the other parts of that complex organ known as the larynx. Let us divide the symptoms into the functional and the laryngoscopic. The connection or antithesis between them will be noticed.

"An initial symptom, quite frequent in tubercular laryngitis, is, without a doubt, cough. This symptom, common to all maladies of the respiratory tract, would have no diagnostic value, except that it is characteristic. On it alone the diagnosis of laryngeal phthisis could never be based. At the beginning, cough puts us on our guard, especially when it is causeless; that is to say, when auscultation of the chest fails to reveal anything abnormal. This cough is always persistent, sometimes violent, hawking, and provoking.

"The physical signs of the chest do not correspond to the tenacity of the cough; it is therefore possible for the larynx to be accused. As regards this cough, the 'hemming' so often described, and which draws attention most often to a possible rhinopharyngitis, may cause us to think at the beginning of tuberculosis, but only after examination of the rhinopharynx has established its integrity. There is a cough, well known at the beginning of tubercular laryngitis, a little dry cough, commencing insidiously, often at the moment when the patient is about to speak, which the individual himself does not notice, but to which his friends attach an importance too often justified by the outcome. The cough may be hacking, followed or not by expectoration, and often accompanied by vomiting. It is certainly right to consider it as a symptom of the beginning of the disease.

"The speaking voice is often altered, dysphonia appears, and the patient who is attacked presents little alteration in his larynx; no ulceration, the cords accurately approximate, and they are very slightly congested; the laryngeal image does not reveal anything by which this profound alteration in the voice can be explained. There is no cough. There will come a time in the disease, however, which will cause us to see that this, too, is an initial form, and oblige us to give a prognosis exceedingly guarded.

"The voice may be eunuchoid. Castex has noted it among the tuberculous. The raucosity of the voice should also recall the statistics which demonstrate the fact that a fifth of the cases of this condition are tubercular. But these three symptoms, dysphonia, raucosity, eunuchoid voice, are also found in other maladies of the larynx; conditions, however, easily diagnosticated by the laryngoscope. If nothing justifies these affections of the voice, one should think of tuberculosis. It is these initial forms, apparently paradoxical, but analogous to that, which we are going to mention under the subject of pulmonary lesion not sufficient to provoke cough in the beginning if the larynx has not been initially affected. The forms that are recognized in the mirror are evidently very numerous. We will mention some: Congestion of the cords, monocorditis, recurrent laryngitis, and a nodular form at the free border of the vocal cords. We do not take into consideration any variety of ulceration, no matter how insignificant, as for the most part the velvety aspect of the cords leads us to think at once of laryngeal phthisis. But this has not appeared at the beginning, and we are only considering initial forms. The symptoms which we are attempting to describe are those suggestive of tuberculosis, and we only say that tuberculosis of the larynx may begin by a nodule, by a congestion, by a monocorditis, etc.

“Congestion of the vocal cords, whose ætiology is difficult to explain, often coincides with slight dysphonia, with cough. This congestion, fugacious, if not tuberculous, disappears with rest, if it is not aggravated by a chronic rhinopharyngitis. In the majority of cases the patient returns. Despite a treatment, properly instituted, the congestion persists; it extends on the cords; it may remain there, or it may reach over the ventricular bands to the arytenoidal apophyses; this is a form of commencing laryngeal phthisis, especially if, after a period of calm, there is found in a patient a new congestion. It is recurrent laryngitis, another form of initial tuberculosis more grave than the first. Against laryngitis of this form treatment is of no avail.

“Another variety of initial tuberculosis is monocorditis. The patient becomes suddenly aphonic; laryngoscopic examination shows a cord perfectly red, congestion of which is evident, not only by the color, but by its altered volume. Contrast with the sound cord is often striking. Movements of the affected cord may be observed, but it is generally paretic. Acute monocorditis should cause us to think that it is an initial form of tuberculosis. This monocorditis often corresponds to the side of the lungs which is afterward or at that time attacked by the bacillus. Certain authors admit that this relation is absolutely constant, and their statistics allow no exception to the rule. On the other hand, Bayle’s theory, setting forth the direct penetration of the tubercular infection, becomes less often justified. It is the lymphatic route which most often produces bacillary infection.

“Tubercular laryngitis may often begin by a nodule situated on the border of the vocal cords. It is important not to confound it with singers’ nodules, these latter being more conical and more rounded. The tuberculous nodule may grow slowly, not ulcerate for a long time; interfering so little with the speaking voice that the patient often refuses any intervention. But the day comes when we see this nodule desquamate, and we may observe the evolution of the tuberculous ulceration which displaces it. We make no mention of the other forms of commencement characterized by a congestion of the entire organ, by œdema of the epiglottis, by a lividity quite characteristic which invades the entire endolaryngeal mucosa, forms most usual for the tubercular involvement of the larynx. A form especially noticeable is that which begins with a sensation of a lump in the throat. It is true that this variety is not observed except in the nervous; it is not, however, to be compared to the globus hystericus. Tuberculous patients, in whom the tuberculous process in the larynx begins with a sensation of a lump in the throat, may be in very good health, but this particular impression is often the first symptom which they observe in a laryngitis, which finally becomes tuberculous. At the moment when the patient complains of this symptom it may happen that laryngoscopic examination fails to detect any lesion. It is useless to add that this form is especially met with in the female. It most nearly resembles that form that begins with a dysphagia that persists to the end; but at the beginning of tubercular laryngitis this dysphagia alone is noted without any other symptoms." So it can be seen that laryngeal phthisis may begin by a variety of symptoms, some common, the others rare. It is needless to insist upon the importance of an early diagnosis.

PALMER.

Ball, James Moores.—On Removal of the Cervical Sympathetic in Glaucoma and Optic-Nerve Atrophy.—*Jour. A. M. A.*, June 2, 1900.

I propose to consider the surgery of the cervical portion of the great sympathetic nerve in certain ocular diseases. European oculists and surgeons have performed sympathectomy for glaucoma and exophthalmic goiter. I have gone further, and in one instance removed the superior cervical ganglion for simple atrophy of the optic nerve. I have performed sympathectomy four times up to July 20, 1899. First the cases will be reported; then the conclusions will be drawn.

CASE I.—EXCISION OF SYMPATHETIC FOR GLAUCOMA ABSOLUTUM.

Mrs. B. S., aged thirty-six, has had pain in and around the right eye for two months, and examination showed vision in this eye reduced to light perception; tension + 3, and the pupil widely dilated. The anterior chamber was shallow, the cornea cloudy and slightly anæsthetic, the media slightly cloudy, still allowing the fundus to be seen. The episcleral vessels were enlarged. Circumcorneal injection was present and the optic nerve cupped. A diagnosis of chronic irritative glaucoma was made. The left eye presents immature cataract, and vision in this eye is 20/70.

Knowing of the flattering results obtained by Jonnesco and others, by excision of the superior cervical ganglion in absolute glaucoma, I explained the operation to the patient, and obtained permission to operate. On May 15, 1899, the patient was anæsthetized, chloroform being employed. An incision four inches in length was made on the right side downward from the mastoid process, extending along the posterior border of the sterno-cleido-mastoid muscle. The external jugular vein was cut and tied. The sterno-cleido-mastoid was then separated from the trapezius muscle, and the spinal accessory nerve was cut. A deep dissection was then made, exposing the carotid sheath. This was opened to enable us to locate the pneumogastric nerve beyond question. The carotid, internal jugular vein, and pneumogastric nerve were then pulled forward, enabling us to see the rectus capitis anticus major muscle, on which the superior cervical ganglion rests. Tearing through the fascia, the ganglion was found and stripped. The ganglion was then cut high up with curved scissors and all its branches severed. About one inch of the trunk of the sympathetic below the ganglion was removed. The wound was closed with interrupted sutures and the neck placed in a plaster cast. The time required for operation was fifteen minutes, and immediately after it was noticed that the right eye was suffused with tears, the right conjunctiva much injected, and the right nostril moist. The intra-ocular tension was + 2. The patient slept well all night, without medicine, being free from pain for the first time in over two months. Tension had steadily decreased to + 1.

On May 16, slight ptosis was noticed on the right side. This symptom is yet present. On May 19 the circumcorneal injection was much less; the conjunctival hyperæmia and lachrymation were still present, while the ptosis was slightly increased and tension was + 1.

At the present date—July 23, 1899—this patient has no pain. The retinal arteries are increased in size. Tension is + 1. Vision has increased from light perception to ability to count fingers at three feet. The conjunctival injection which followed the operation has disappeared; the optic nerve has a color more approaching the normal. The ptosis is less.

This was the first sympathectomy made in America for glaucoma.

CASE II.—DOUBLE SYMPATHECTOMY FOR GLAUCOMA SIMPLEX.

Miss M. E., a German, aged forty-three, was sent to me on June 14, 1899. For two years sight had been failing, until at this time vision was as follows: R. E. = 0; L. E = light perception. Tension was + 3. Both optic nerves showed marked cupping of the disk; the vessels were pushed to the nasal side. She stated that she had never had pain in the eyes, and had not consulted an ophthalmic surgeon.

I advised her to submit to an excision of the left superior cervical ganglion; she consented, and on June 15 the operation was performed by myself, assisted by Dr. E. C. Renaud, at St. Joseph’s Sanatorium, in the presence of Drs. J. C. Murphy, A. R. Kieffer, and S. A. Grantham. The operation was difficult, owing to the abnormal position of the vagus nerve. This was outside of and external to the carotid sheath, and was much smaller than normal; it was not larger in diameter than the head of a pin. It was identified by irritating it and watching the effect on the heart. The superior cervical ganglion was removed and one-half inch of the trunk of the sympathetic below. Shortly after the operation there were lachrymation, ocular congestion, and contraction of the pupil on the corresponding side. On the second day she counted fingers at 2½, and on the third at 3½ feet. Slight ptosis was present.

She left the hospital on the eighth day. At this time she counted fingers at four feet. There was only slight, if any, reduction of tension during the eight days she was in the hospital. In counting fingers she saw with the nasal side of the retina—temporal field. I did not see her again until June 30, and she was then counting fingers at five feet. Tension on that day was normal. She had light perception in the right eye.

On July 16 I excised the right superior cervical ganglion without difficulty, and on July 7 she counted fingers at seven feet with the left eye, and could see the hand at four inches with the right. I examined her on July 20, when vision remained the same, the tension of the right eye was + 1, and of the left + 2. She was well pleased to have the small amount of vision she possessed.

CASE III.—SYMPATHECTOMY FOR OPTIC-NERVE ATROPHY.

T. J., aged forty-six, an inmate of the St. Louis City Hospital, a laborer, was admitted on account of blindness. There was no history of syphilis, rheumatism, nor any systemic disease. The patient was of limited mentality. No history of his family could be obtained. He claimed to have had good health all his life, with the exception of an attack of malarial fever several years ago. The patient had been a moderate drinker of alcoholic beverages. In appearance he was robust, and he complained only of loss of vision which, in the left eye, had been failing for eleven months, in the right for seventeen weeks, according to his statement. Until seventeen weeks before this he could see enough with the right eye to get around. Since then vision had steadily declined until he had light perception only—and this only apparent when light was concentrated on the eye by the ophthalmoscopic mirror. Vision of the left eye = 0.

The pupils were widely dilated. The ophthalmoscope showed, in the right eye, a white disk, particularly on the temporal side; the arteries slightly reduced in caliber, veins normal. There was shallow, atrophic cupping of the nerve head. The retina and choroid were normal, the vitreous and lens clear. The left eye showed a disk of a dead white color throughout the whole area, arteries very small, atrophic excavation pronounced, veins reduced in caliber, and choroid normal. The macula was not visible in this eye, owing to the much-reduced blood-supply. The vitreous and lens were clear. Vision was as follows: R. E. = perception of concentrated light. L. E. = 0.

Diagnosis.—R. E. = optic-nerve atrophy. L. E. = complete atrophy of optic nerve and retina.

Treatment: Resection of the right superior cervical ganglion of the sympathetic was done. The operation was followed by conjunctival congestion, lachrymation and contraction of the pupil, slight ptosis and hypotonia.

No appreciable change in the patient’s vision followed, and ophthalmoscopic examination made two weeks after operation showed no change in the appearance of the fundus, except that a cilioretinal artery in the upper part of the disk had doubled in caliber.

So far as I know, Case III. is the first instance in the history of medicine of an excision of the superior cervical ganglion, or of any part of the sympathetic system, for the relief of optic-nerve atrophy. Although the operation was not of benefit in this particular instance, yet I am not willing to concede that it will prove valueless in cases of non-inflammatory atrophy in which vision is not entirely lost. In truth, I expect it to prove beneficial in such cases, sufficiently often to justify the procedure.

I was led to make this experimental operation for several reasons: 1. The use of glonoin is often followed by an improvement in vision in cases of simple atrophy of the optic nerve. 2. Glonoin enlarges the retinal vessels, as has been proved by ophthalmoscopic examination. 3. There is no question that in glaucoma simplex—a disease in which there is an atrophy of the optic nerve—improvement in vision follows sympathectomy. 4. Excision of the cervical sympathetic is followed by an increase in the blood-supply of the orbital contents.

PATHOLOGIC CHANGES IN THE EXCISED GANGLIA.

The microscopic examination of three of the excised ganglia was made by my friend, Dr. Carl Fisch, of St. Louis. The specimens were those from Cases I., II., and III. Of the two ganglia removed from Case II. only the first one—the left—was examined.

Transverse and longitudinal sections of the three specimens were studied microscopically, by means of a great number of different staining methods. Owing to the method by which the ganglia had been preserved—weak formalin solution—the employment of the Golgi—Marchi—and the more delicate Nissl stains was rendered impossible. In general it may be said that the pathologic changes found were the same in the three cases, although a little less pronounced in No. 2 than in 1 and 3.

Most striking of all was a very marked hyperplasia of the connective tissue, which in some places resulted in dividing up the ganglion into small groups of nervous elements separated by broad bands of fibrous elements. The walls of the vascular structures showed decided sclerosis; the connective-tissue sheaths of the ganglionic cells were much increased in thickness. In Case I. small foci of round-cell infiltration were seen in this hyperplastic growth, of an inflammatory character. No plasma nor mast cells could be demonstrated.

The ganglionic cells were markedly pigmented. Together with a number of cells normal to all appearance there were great numbers showing different stages of degeneration. As a rule the nucleus, besides having lost part of its peculiar staining property, had assumed the parietal position; the nucleus was reduced in size or even missing in a large percentage of the cells. While in some cells the chromatic elements were well preserved, in others the process of chromatorhexis and chromatolysis could be followed up through all of its stages. Only comparatively few cells were seen showing the normal dendriform processes; very often the processes were short, ending bluntly, or they had even disappeared altogether. The general peripheral network of processes was much reduced in volume and compressed by the pressure of the connective-tissue formation. Only very few medullated fibers were seen. Unfortunately it was impossible to study their structure with the Marchi method.

The general pathologic aspect was that of a decided sclerosis, originating in inflammatory processes going on in, and starting out from, the walls of the vascular structures. The changes of the nervous elements were most likely not idiopathic, but due to pressure and inhibited nutrition.

The plates accompanying this paper have been made from drawings of sections of superior cervical ganglia.

TECHNIQUE OF THE OPERATION.

The ordinary precautions for surgical cleanliness are to be observed, and general anæsthesia employed. The incision should be made along the posterior border of the sterno-cleidomastoid muscle, starting at the mastoid process and running downward to within an inch of the clavicle. The sternomastoid is separated from the adjacent muscles, the spinal accessory nerve cut, and the carotid sheath reached. This dissection is made with the fingers. The carotid sheath should always be opened in order to locate the pneumogastric nerve. I consider this very important because: 1. The nerve is sometimes outside the sheath, as happened in my second case, in which the pneumogastric was much atrophied and was external to the sheath. 2. Differentiation of the cervical sympathetic from the vagus is sometimes difficult. Often, in operating on the cadaver, I have found both nerves inclosed in the same fascia. It is needless to say that excision of the vagus instead of the sympathetic would not only defeat the object of the operation, but would add a serious complication. Differentiation of these nerves after opening the carotid sheath is not usually difficult, for in working upward the operator comes upon the ganglionic expansion of the sympathetic. The ganglion is seized with forceps and stripped. Its branches are cut first, then the cord passing below is severed, and lastly the ganglion is cut above, as high as possible. It is best to use curved scissors and to have the finger under the ganglion while traction is made, thus cutting on the finger and avoiding injury to the underlying structures.

If the middle ganglion is to be removed, it will be best to excise it first and then work upward. If the entire chain of the sympathetic is to be removed, as is done for epilepsy, and as is now advised in exophthalmic goiter by Jonnesco, the operation is one of great difficulty, owing to the location of the inferior ganglion. This is situated near the neck of the first rib. One of my friends, who is a skillful surgeon, in removing this ganglion ruptured the vertebral artery near its origin and was obliged to tie the subclavian to check the hemorrhage. After the latter has ceased the wound is closed with superficial sutures. The hemorrhage in removal of the superior ganglion is usually trifling, only a few small vessels being cut. The external jugular vein was cut in my first case, but not in the others. The patient leaves the hospital on the eighth or ninth day.

Jonnesco's method, according to his latest communication on the subject, is different. He always employs the premastoid route where only the superior ganglion is to be removed, reserving the postmastoid for the excision of the entire chain. The carotid sheath is split, the internal jugular vein and sternomastoid drawn outward by a retractor; a second retractor draws the vagus and internal carotid inward. In the space made the superior ganglion is found. The deep vertebral fascia is opened, all the branches of the ganglion isolated and cut by blunt, curved scissors; when this has been done the ganglion is attached only by nerve strands above, a strong pull is made, and the ganglion gives way. The excision is then completed by cutting the inferior strands. In closing the wound, he uses both deep and superficial sutures.

He mentions a transient dysphagia and pain in the cranio-mandibular joint as occurring after this operation.

EFFECTS OF EXCISION OF SUPERIOR CERVICAL GANGLION.

The effects of removal of this ganglion are immediate and remote: The immediate are relief of pain, lachrymation and conjunctival injection, together with a discharge from the corresponding nostril, unilateral sweating, and contraction of the pupil. Often there is an immediate reduction in intra-ocular tension. These effects are noted within five minutes after the excision.

The remote effects are ptosis, which appears on the third or fourth day, improvement of vision, and in some instances a tardy contraction of the pupil and a tardy reduction of the intra-ocular tension. To these there must also be added a slight sinking of the eyeball into the orbit, and a feeling of heaviness in the head. What I have just written applies particularly to cases of glaucoma.

In exophthalmic goiter, after the excision of the ganglia, the exophthalmus and tachycardia are said to improve almost immediately and a reduction of the goiter soon follows.

Although Jonnesco speaks of the immediate reduction of the intra-ocular tension, yet this does not always occur. In my second case, at the end of eight days the tension was + 2. On the sixteenth day the tension was normal. In my first case reduction of the tension was immediate. The relief from pain in the first case was immediate and lasting. This patient had not been free from pain for two months previously. The slight ptosis following sympathectomy is to be attributed to paralysis of Müller’s muscle. Sinking of the eyeball is no doubt due to paralysis of the unstriped peribulbar fibers found in Tenon’s capsule. Contraction of the pupil is usually an immediate result; it may, however, appear tardily. Thus in my first case the pupil was unchanged until the fourth day after the operation; and it did not become at any time as markedly contracted as in the other two patients. In the third case—that of optic-nerve atrophy—the pupil was markedly contracted within five minutes after the excision.

The lachrymation, conjunctival injection, and nasal moisture are transient symptoms which are usually absent after the first day.

In this connection it is interesting to note that Mr. Jonathan Hutchinson, as early as 1866, recognized many of the ocular symptoms of paralysis of the cervical sympathetic, and wrote a paper thereon.

HOW DOES EXCISION OF THE CERVICAL SYMPATHETIC REDUCE INTRA-OCULAR TENSION?

This is a question difficult to answer—difficult for the reason that we are not sufficiently acquainted with the physiology of the production of aqueous humor under normal surroundings. Panas and Duvigneaud have assumed rightfully that “If the nervous mechanism of intra-ocular secretion or, to speak without hypothesis, the action of the nervous system on intra-ocular tension can be known, the pathology of glaucoma will be cleared up, iridectomy will be explained, and perhaps a new and scientific basis for the treatment of glaucoma will be established.” Many observers have sought to solve the problem. Donders attributed the hypertension to a neuro-secretory cause and believed the trigeminus to be the agent of excessive secretion. He held that section of the trigeminus should relieve intra-ocular tension, while section of the cervical sympathetic could have no particular influence.

His views were overthrown by experiments made by Wegner in 1866, on rabbits. By means of manometers placed in the anterior chamber, he sought to record variations in the intraocular tension. He proved to his own satisfaction that the trigeminus takes no part, while section of the cervical sympathetic produces hypotonia, and irritation of its upper end and causes hypertonia. He held that section of the cervical sympathetic enlarges the blood vessels of the eye; the blood then flows under reduced pressure, and intra-ocular secretion is lessened. Almost identical results were obtained by Adamück—1866–68—who experimented on cats.

Von Hippel and Gruenhagen believed that the cervical sympathetic contains vasoconstrictor fibers for the eye. Their experiments were made on cats and dogs. They found that irritation of the upper end of the cervical sympathetic causes in the cat hypertonia, while its extirpation increases intra-ocular tension. While, according to Wegner, the hypertonic action proceeds from the enlargement of vessels caused by cutting the cervical sympathetic, and the contraction of the blood vessels caused by the irritation of the nerve causes a hypertonic action, the contrary view is held by Adamück, Von Hippel, and Gruenhagen.

However this may be, there is no doubt that the trigeminus plays no great part in the production of ocular tension. Furthermore, the inefficiency of Bedal’s operation—stretching the nasal nerve—is explained by the fact that it is the cervical sympathetic, and not the trigeminus, which influences intra-ocular tension.

Jonnesco believes that the ocular sympathetic fibers from the brain and spinal cord pass through the superior cervical ganglion; permanent or intermittent irritation of these is accompanied by dilatation of the pupil, narrowing of the small intra-ocular arteries, contraction of the peribulbar muscular fibers, and probably an increased action of the elements which produce the aqueous humor. “As a matter of fact,” says Jonnesco, “any increase of the blood pressure will produce a permanent or intermittent narrowing of the arteries and cause the extravasation and increase in aqueous humor; then it is probable, although not definitely settled, that a permanent or intermittent irritation of the excito-secretory fibers is followed by an increase in the secretion of aqueous humor; the permanent or intermittent dilatation of the pupil pushes the iris into the iris-angle, closes the canals of the filtration zone, and hinders or prolongs the exit of aqueous humor from the eye; the permanent or intermittent contraction of the unstriped peribulbar muscular fibers closes the efferent veins of the eyeball, and hinders the venous circulation of the eye—hence the dilatation of the intra-ocular veins."

He holds that excision of the superior cervical ganglion destroys all vasoconstrictor fibers of the eye. The arteries relax, the blood pressure is lowered, and extravasation is reduced. This operation destroys the excito-secretory fibers, thus limiting the amount of aqueous produced. The fibers which dilate the iris are destroyed, hence the contraction of the pupil reopens the iris-angle and removes the obstacle to the outflow of aqueous. The nerve-fibers supplying the unstriped muscular apparatus contained in Tenon's capsule are destroyed, hence the pressure on the efferent veins is removed and ocular circulation is reestablished.

Jonnesco believes that the starting-point of the nervous derangement producing glaucoma is central: "When one removes the ganglion the point of origin of the influence will not be removed, but the communication between this center and the eyeball is destroyed."

Regardless of the differing views of physiologists concerning the mechanism of the reduction of ocular tension, based on experiments made on the lower animals, there can be no difference of opinion concerning the effect of excision of the superior cervical ganglion in the human subject. The operations made by Jonnesco and others on the Continent, and by myself in America, prove that removal of the superior cervical ganglion causes a marked reduction of intra-ocular tension in glaucomatous cases. That the same effect occurs in eyes with normal tension is evident from my third operation—that done for optic-nerve atrophy.

EXTENT OF SYMPATHECTOMY IN DIFFERENT DISEASES.

Up to the present time excision of the cervical sympathetic has been performed for the following diseases: epilepsy, exophthalmic goiter, glaucoma, and optic-nerve atrophy. The question naturally arises: How extensive an operation is necessary in these affections? This I will attempt to answer:

In epilepsy it is necessary to excise the entire cervical chain on both sides for the reason that, according to Jonnesco's theory, it is necessary to convert a state of cerebral anæmia—which he assumes is the condition in epilepsy—into one of cerebral hyperæmia. Since the carotid plexus is formed by branches from the superior ganglion, and the vertebral plexus arises from branches which have their origin in the inferior cervical ganglion, it is evident that the entire cervical sympathetic must be removed.

In exophthalmic goiter, although Jonnesco in his first operation excised only the superior and middle ganglia, he now believes it necessary to remove the inferior as well, for this reason: from the superior ganglion the ocular fibers arise; from the inferior the vasodilator, cardiac-accelerator, and, probably, the secretory nerves of the thyroid gland. If eye, thyroid, and cardiac symptoms are to be relieved the entire chain must be excised.

In glaucoma removal of the superior ganglion alone is necessary. All of the sympathetic fibers of the eye, with the exception of those which pass directly from the cerebrum by way of the trigeminus, are connected with the superior ganglion.

In optic-nerve atrophy, if it should be proved that noninflammatory atrophy of the optic nerve can be improved by sympathectomy, removal of the superior ganglion alone will be necessary, for reasons already given.

If the glaucoma is unilateral, it is necessary to remove only the corresponding ganglion.

HISTORY OF SYMPATHECTOMY.

In 1889 Alexander of Edinburgh resected the superior ganglion on both sides. In 1892 Jacksh resected the vertebral plexus and cut the cord connecting the middle and inferior ganglion. The third operator was Kummel, who excised the superior ganglion on one side only. In 1893 Bojdanik made a bilateral resection of the middle ganglion. In 1896 Jaboulay made a bilateral section of the sympathetic cord, above and below the middle ganglion. These operations were all made for epilepsy.

In regard to exophthalmic goiter, Jaboulay made a simple section of the sympathetic early in 1896. In September of the same year Jonnesco excised the superior and middle ganglia.

Jonnesco was the first, in 1896, to do a bilateral resection of all three cervical ganglia, though it is claimed by a Polish surgeon, Baracz, that he proposed the same in 1893. To Professor Jonnesco furthermore belongs the credit of having first excised the superior ganglion for glaucoma in September, 1897.

Ball of St. Louis was the first to remove the superior cervical ganglion for optic-nerve atrophy. The date of this operation was June 24, 1899.

Terrier, Guillemain, and Malherbe, in their “Chirurgie du Cou,” 1898, were among the first to give the surgery of the sympathetic a place in a text-book.

Among those who have operated on the cervical sympathetic for the relief either of glaucoma or exophthalmic goiter, or both, are Abadie, Réclus, Gerard-Marchant, Chauffand and Quénu, Jeunet, Bled, Ball, Renaud, and Bartlett.

Panas is opposed to sympathectomy in glaucoma. He reports seeing a patient in whom, three months after the operation, vision was still declining.

François-Frank, at a meeting of the Paris Academy of Medicine, held May 22, 1899, spoke of the effect of sympathectomy on the circulation of the thyroid gland, brain, and eyes, and on the heart. He believes that the operation can easily produce good results.

Doyon has described the trophic changes produced in the rabbit by excision of the cervical sympathetic.

CONCLUSIONS.

From a study of the cases of sympathectomy made by Jonnesco and others, and from the observation of my own cases, I offer these conclusions:

1. Excision of the superior cervical ganglion is a most valuable procedure in glaucoma.

2. It is of more value in glaucoma simplex than in inflammatory glaucoma.

3. In inflammatory glaucoma, on which iridectomy has been done without benefit, excision of the superior cervical ganglion should certainly be tried.

4. In cases of absolute glaucoma with pain, sympathectomy is to be tried before resorting to any operation on the eyeball.

5. In cases of simple optic-nerve atrophy, sympathectomy may possibly be beneficial if done before vision is entirely lost.

6. In cases of exophthalmic goiter, which do not improve under hygienic medicinal and electric treatment, excision of the cervical sympathetic on both sides is to be advised.

7. In unilateral glaucoma excision of the sympathetic ganglion is to be done only on the corresponding side.

8. In the hands of a careful operator, excision of the superior and middle ganglia is a safe operation, but removal of the inferior ganglion can be done safely only by the most skillful surgeons.

9. The postmastoid route is to be preferred in excision of any part, or all of the cervical sympathetic.

10. The fact that glaucoma is improved by sympathectomy and the finding of pathologic changes in the excised ganglia suggest the conclusion that this affection is due either to a permanent irritation of the cervical sympathetic, or to an irritation located elsewhere and transmitted by means of the cervical sympathetic.

I wish to extend my thanks to Drs. E. C. Renaud and Willard Bartlett for valuable assistance in the preparation of this paper; to Dr. Carl Fisch for the pathologic report.

DEADY.

Heath, M. D., Charles.—A Case of Sinuses in the Vault of the Naso-pharynx.—*The Jour. of Lar., Rhin. and Otol.*, May, 1900.

Case shown at the Lar. Soc. of London:

A woman, æt. thirty-one years, had suffered several years with discomfort in nose, throat, and mouth, with dyspepsia. Mucosa of nares and pharynx markedly atrophied. Atrophied condition made post-rhinoscopy easy. “The eustachian eminences were seen to be enormous, filling the fossæ of Rosenmüller, and reaching nearly to the pharyngeal roof. Just behind the upper edges of the choanæ, on each side, there appeared a transverse elliptical opening, which was about half an inch long and a fifth of an inch across at the widest part on the left side, and slightly less in each dimension on the right; a probe apparently extends about a quarter of an inch into the cavity.” In the discussion following some thought them to be small recesses formed by cicatricial tissues, other formed by peculiar distribution of adenoid tissue, and still genuine sinuses.

Palmer.

Roughton, B. S. (Lond.), F. R. C. S., Edmund W.—The Diagnosis and Treatment of Chronic Purulent Nasal Discharges.—*The Jour. of Lar., Rhin. and Otol.*, May, 1900.

We ascertain by interrogation, (*a*) if the discharge is purulent or muco-purulent; (*b*) whether it is unilateral or bilateral; (*c*) whether it is continuous, intermittent, or influenced by change of posture; (*d*) if there is offensive smell perceived by the patient or by others; (*e*) pain is not usually complained of unless there is obstruction to drainage, and consequently retention of pus under pressure. Unilateral discharge suggests a foreign body in a child or sinus involvement in an adult. If it is intermittent or influenced by position probably originates in a sinus. Subjective fetor suggests sinusitis; while objective fetor, ozena; and combined subjective and objective is the rule in syphilitic necrosis. Location of pain is of very little, if any, use in diagnosis.

Rhinoscopy.—Attention directed to (*a*) situation of the pus; (*b*) polypi; (*c*) atrophy of the mucous membrane; (*d*) crusts; (*e*) ulcerations; (*f*) adenoids; (*g*) nasal obstruction; (*h*) foreign bodies. Under (*a*) beside usual cleansing of nasal cavities and reexamination to ascertain situation, he recommends "tamponading," *i. e.*, by blocking up first one part, then another, with pledgets of wool, and noticing whence the discharge reappears. (*e*) Ulceration may be syphilitic, simple, tubercular, or lupoid in origin. "It must not be forgotten that a perforation" of the septum "may be entirely the work of a misused finger-nail." (*g*) The normal mucoid discharge damned up by nasal obstruction frequently becomes purulent.

Special methods of diagnosis as follows are mentioned: transillumination; examination of upper teeth; catheterization of the ostium, maxillares, naso-frontal canal, and outlet of the sphenoidal sinuses; external examination of antrum and frontal sinuses; exploratory puncture of the antrum through the inferior meatus, alveolar process, or canine fossa. Diagnosis of ethmoiditis is principally by exclusion.

Treatment of the accessory sinuses may be summed up under the following indications: (*a*) removal of the cause; (*b*) evacuation and drainage of pus; (c) antiseptic irrigation; (d) removal of morbid material, when present. PALMER.

Williamson, R. T.—Remarks on the Diagnosis and Prognosis in One Hundred Cases of Double Optic Neuritis with Headache. —*Lancet*, May 12, 1900.

The detection of optic neuritis is of the greatest importance in the diagnosis of cerebral affections. Nevertheless, in certain cases of double optic neuritis with headache considerable caution is necessary before coming to a conclusion as to the exact nature of the disease. Though these two symptoms are present in the majority of cases of brain tumor and are so frequently due to this cause they are also met with in other diseases. In some cases of granular kidney, for example, the patient comes under treatment for headache and failure of vision; and ophthalmoscopic examination may reveal intense optic neuritis like that of cerebral tumor (neuritic form of albuminuric retinitis). At first the symptoms appear to indicate cerebral tumor, but a careful examination of the urine and cardio-vascular system will clearly reveal the cause. Limited space does not permit an enumeration of all the causes of double optic neuritis with headache. The results of the examination of one hundred cases presenting these two symptoms reveal, however, several points of interest. Most of these cases have been seen by us conjointly; some were seen separately, whilst others (in Groups I. to VIII.) were examined by one of us (R. T. W.) whilst holding the post of medical registrar at the Manchester Royal Infirmary. For permission to include the latter amongst our cases we are indebted to the medical board of that hospital.

With respect to the diagnosis and termination these one hundred cases may be grouped as follows:

I. Brain tumor, verified by necropsy, 27.

II. Cases terminating fatally; probably, majority due to brain tumor; but no necropsy obtained, 27.

III. General symptoms of brain tumor; but necropsy revealed distention of the ventricles of the brain with fluid; no tumor (serous meningitis of ventricles), 2.

IV. Cerebral abscess (fatal), 3.

V. Tuberculous meningitis (fatal), 2.

VI. Chronic interstitial nephritis; neuritic form of albuminuric retinitis (fatal), 3.

VII. Toxic conditions and blood diseases: Chronic lead poisoning, 3. Ulcerative endocarditis, (fatal), 1. Purpura hemorrhagica (fatal), 1. Henoch’s purpura (fatal), 1. Chlorosis with cerebral symptoms (recovery), 3.

VIII. Headache and double optic neuritis (without localizing symptoms); probably syphilitic (recovery, with blindness, 2; with impaired vision, 4), 6.

IX. Headache and double optic neuritis (without localizing symptoms); no evidence of syphilis; duration six and two and a quarter years, respectively. Termination still uncertain, 2.

X. Headache and double optic neuritis (without localizing symptoms); no evidence of syphilis; recovery with blindness, 8; with impaired vision, 3; with good vision, 8, 19.

The following are brief abstracts of the notes of the cases in Group X. which have come under our observation and which we have followed for a considerable period of time. The number of years during which each case has been followed is given in parentheses after the brief note.

1. A boy, aged ten years. Double optic neuritis, headache, and vomiting; slight internal strabismus of the left eye. Recovery with normal vision. (Seven years.)

2. A young woman, aged seventeen years. Headache, vomiting, and double optic neuritis. Recovery, but with impaired vision in one eye and blindness in the other. (Five and a half years.)

3. A young woman, aged eighteen years. Double optic neuritis, headache, and vomiting; several epileptic fits. Recovery, with useful vision in one eye; vision in the other is very defective. (Seven years.)

4. A young woman, aged eighteen years. Double optic neuritis, headache, and vomiting. Recovery, but complete blindness followed. (Four years.)

5. A young woman, aged nineteen years. Double optic neuritis, headache, and vomiting. Recovery with good vision. (Three years.)

6. A man, aged twenty years. Double optic neuritis, headache, and vomiting. Recovery, but complete blindness followed. (Two and a half

years.)

7. A girl, aged ten years. Double optic neuritis, headache, and vomiting; knee-jerks were absent. Recovery, but complete blindness followed. (Three years.)

8. A boy at the age of thirteen years had double optic neuritis and headache; recovery ensued. At the age of fifteen years he had a return of headache and double optic neuritis; also vomiting. At a later date there was partial anæsthesia in the distribution of the right fifth cranial nerve; the right cornea was opaque; there was complete blindness in both eyes. Partial anæsthesia of the face and blindness remained, but otherwise the patient recovered and felt quite well nine months after the second attack.

9. A girl, aged sixteen years. Double optic neuritis, headache, vomiting; slight internal strabismus of the right eye. Recovery with normal vision. (Four and a half years.)

10. A boy, aged ten years. Double optic neuritis, headache, and vomiting. The head had increased in size. Recovery, but with complete blindness. (Three years.)

11. A woman, aged nineteen years. Double optic neuritis, headache, and vomiting. Recovery with normal vision. (Three years.)

12. A boy, aged fifteen years. Double optic neuritis, headache, and vomiting. Recovery. (Two and a half years.)

13. A woman, aged twenty-one years. Double optic neuritis, much swelling of the disks, headache, and vomiting. Complete recovery with normal vision. (Four and three-quarter years.)

14. A girl, aged fifteen years. Double optic neuritis, headache, and vomiting. Recovery with good vision. (Five and a quarter years.)

15. A boy, aged twelve years. Double optic neuritis, headache, and vomiting; slight internal strabismus of the right eye. Recovery with good vision. (Four years.)

16. A man, aged forty years. Double optic neuritis, headache, and vomiting. Recovery, but with complete blindness. (Eighteen months.)

17. A youth, aged seventeen years. Double optic neuritis, headache, and vomiting. Recovery, but with total blindness. (Five years.)

18. A woman, aged twenty-two years. Double optic neuritis, headache, and vomiting. Recovery, but with total blindness. (Two and three-quarter years.)

19. A girl, aged fourteen years. Double optic neuritis, headache, and vomiting; internal strabismus (left) for fourteen days. Recovery, with normal vision. (Two years.)

The following are brief notes of the cases in Group IX.:

20. A girl, aged twelve years. Headache, vomiting, and double optic neuritis in December, 1893. Recovery in twelve months, but vision was much impaired. She remained well with the exception of occasional headache until December, 1899. Then the severe headache returned. She became ataxic and optic neuritis reappeared. In April, 1900, the headache was much less and the patient felt much better, but she was completely blind. (Six and a half years.)

21. A young woman, aged seventeen years. Headache, vomiting, and double optic neuritis. Vision was impaired. Vomiting ceased; the headache continued for over two years, but recently disappeared after lumbar puncture. (Two and a quarter years.)

In all cases of double optic neuritis a systematic and careful examination of the patients should be made. The urine and cardio-vascular system should be examined for signs of chronic interstitial nephritis; the gums should be examined for the lead line and other indications of lead poisoning should be sought for; the question of chlorosis or other “blood disease” should be considered; and the ears should be examined for signs of otitis. But when all these conditions have been excluded and when the symptoms are apparently due to a cerebral affection, there is one group of cases in which localizing brain symptoms are absent and in which the chief indications of disease are headache, double optic neuritis, and often vomiting. In most of these cases syphilis can be also excluded. A diagnosis of brain tumor is given, and the growth is thought to be situated in some region in which the localizing symptoms are at first indefinite—cerebellum, temporo-sphenoidal lobe, or prefrontal region. Such a diagnosis often proves to be correct. Localizing symptoms may develop later and a necropsy may show the accuracy of the opinion expressed. But sometimes, to the surprise of the medical man, a fatal termination does not occur; the symptoms sometimes disappear and the patient recovers, though very often impairment or loss of vision remains. The patient may continue in good health for years or for a lifetime afterwards. Most medical men who have paid much attention to cerebral diseases will have met with a case or cases of this kind. The chief object of our article is to call attention to this class of cases and to indicate the frequency of their occurrence. Nineteen out of one hundred cases of double optic neuritis with headache in the table just given could (after careful examination) be placed in this group (X.).

What is the cause of the symptoms in this group of cases? Possibly in some cases the symptoms are caused by a non-malignant tumor (or tuberculous mass) which ceases to extend and becomes quiescent and encapsuled. One of us has recorded a case in which symptoms of cerebral tumor (including Jacksonian epilepsy and hemiplegia) gradually subsided and temporary recovery ensued; but three years later symptoms of cerebellar tumor developed and death occurred. The necropsy revealed a recent large tuberculous mass in the cerebellum and an old capsuled tuberculous mass just beneath the motor cortex in the right cerebral hemisphere. The latter had evidently been the cause of the early cerebral symptoms from which the patient had recovered. An instructive case has been recorded by Dr. T. K. Monro of Glasgow. The patient, at the age of sixteen years, suffered from severe headache with failure of vision which passed on to complete blindness. For thirty-three years he was an inmate of a blind asylum, ophthalmoscopic examination showing double optic atrophy. He died at the age of sixty-three years, from cancer of the stomach, and the post-mortem examination also revealed a large myxomatous tumor in the left half of the cerebellum. In all probability the early cerebral symptoms had been associated with optic neuritis which had passed on to optic atrophy and the cause had been the myxoma in the cerebellum which had remained quiescent for forty-six years.

In some cases of double optic neuritis with headache and general cerebral symptoms, when recovery occurs the cause is probably distention of the ventricles of the brain with fluid—serous meningitis of the ventricles (Quincke). This condition was present at the necropsy, and no tumor growth could be found in two out of the one hundred cases tabulated. It is probable that a number of the cases in which a diagnosis of cerebral tumor has been made, but in which recovery has occurred, have been due to this condition—serous meningitis of the ventricles. Probably the two cases in Group IX. and possibly some of the cases in Group II., in which death did not occur for several years after the onset of symptoms, were of this nature. Other cases which recover may be due to a basal meningitis.

The table given above is instructive both as regards the diagnosis and prognosis in cases of double optic neuritis with headache. It shows the necessity for careful examination before giving either a diagnosis or prognosis, and the clinical group of cases No. X. ought always to be borne in mind whenever the diagnosis is obscure and localizing symptoms are absent.

There are two other points to which we would draw attention. In ten out of the one hundred cases the patient recovered completely from the headache and general cerebral symptoms and regained perfect health, but the optic neuritis was followed by atrophy and complete blindness. In the face of this terrible termination we cannot help thinking that simple trephining of the skull and the removal of bone, without any interference with the brain, as suggested and practiced by Mr. Victor Horsley for the relief of optic neuritis and pressure symptoms, is a method of treatment worthy of more frequent trial when vision is failing markedly. Dr. James Taylor has published cases which appear to show that this method of treatment may be of service in checking the optic neuritis and failure of vision. In the class of cases in Group X. if there should be a suspicion that the symptoms may be due to serous meningitis of the ventricles, lumbar puncture appears to be worthy of trial, since several cerebral cases are now on record in which this treatment appears to have been of great service, and in which the cause of the cerebral symptoms was probably that just mentioned.

DEADY.

Murrell, W.—A Case of Double Optic Neuritis from Serous Effusion (Quincke's Disease).—*Lancet*, April 28, 1900.

A schoolboy, aged seven years, was admitted into Westminster Hospital on January 28, 1900, the only history obtainable being that on the previous morning he had been brought home in a "fit," which lasted the greater part of the day. On admission he was perfectly sensible and talked freely, but on being put to bed he passed into a condition of semi-consciousness which lasted for many days. He took no notice when spoken to, and remained absolutely mute. The face and upper extremities exhibited choreiform movements of a slow and coarse type. These movements were apparently purposive in character, and at times he endeavored to clutch at objects within his reach. Sometimes the arms were widely extended, and then slowly flexed, as if performing the act of embracing. Sometimes the movements conveyed the idea that he was feebly endeavoring to strike those around him. There was no paralysis of the face or of the muscles of the limbs. The movements were, as a rule, bilateral, although sometimes the facial movements were unilateral, but not always on the same side. There was no rigidity of the muscles, retraction of the head, or opisthotonos. There was nothing to indicate that the patient suffered from headache, although at times the brows were contracted and the face wore a worried and anxious appearance. The bowels were open twice a day and urine and fæces were passed in bed. The motions were normal in character. The patient was unable to swallow, and had to be fed by the nasal tube. There was no nystagmus, the pupils were normal in size and contracted well to light. There was well-marked double optic neuritis. The temperature was 99.8° F., and the pulse was 108. There was no tenderness or swelling of the joints, and there was no rash on the skin. No tache cérébrale could be obtained. There was a little cough, but there was no expectoration. The breath and heart-sounds were normal. The urine was acid, had a specific gravity of 1018, and contained neither albumen nor sugar. The spleen was not enlarged. The patient showed no signs of anæmia, but the blood was not examined. There was no wasting of the muscles, and the knee-jerks were present, although somewhat sluggish. The tongue was clean, and presented no sign of having been bitten. The patient would not protrude it voluntarily, and it had to be examined with the spatula.

The condition of the patient remained practically unchanged for twelve days. The highest temperature recorded was on the second day, when it reached 100°; on the following day it was 99.8°, and from that time onward it was normal. The double optic neuritis continued, and the disks were observed to be getting paler. On February 13 (the seventeenth day of the illness) the patient was much more sensible, and recognized his mother, putting his arms round her neck. He was still unable to talk, although apparently he endeavored to do so, from time to time uttering a few unintelligible words. On being asked if he would like an orange he nodded his head, and he showed some signs of interest in a watch which was shown to him. The incontinence of urine and fæces continued, but food was taken with less difficulty. The movements gradually subsided. On the 17th the patient could say his own name, but beyond that could utter only inarticulate sounds, and failed to recognize letters or words, either written or printed. On the 20th he was able to speak plainly, although incoherently. He endeavored to get out of bed, and during the night was so noisy that he had to be removed from the ward. Urine and fæces were still passed under him. On the 22d he was quieter, and for the first time indicated that he wanted the bed-pan. The optic neuritis was less marked. On March 1 the patient was able to get up, and seemed to be quite well. On the 8th the following note was furnished by Mr. G. Hartridge, who had frequently examined his eyes during the course of his illness: “Pupils five millimeters each. React well to light, to convergence, accommodation, and consensually. Right vision $\frac{6}{6}$, left vision $\frac{6}{9}$. Right disk getting white; not much swelling of the disk; edges clearing. Retinal vessels, specially veins, very full and tortuous. Left disk pale (less so than right), dim; edges blurred.” The only medical treatment adopted was the administration for a few days of 15 minims of liquor arsenicalis three times a day.

DEADY.

Stephenson, Sydney.—Concussion of the Retina.—*Brit. Med. Jour., January, 1900.*

Several years have elapsed since Dr. R. Berlin described a series of cases in which he had observed a peculiar retinal change after the eye had been struck with a blunt object, as, for example, a stick or a stone. Under those circumstances he noticed a cloudiness of portions of the retina, not involving the retinal blood vessels. The milky appearance reached its height in twenty-four to thirty-six hours, and disappeared in two or three days. Berlin pointed out that the rapidity with which the cloudiness developed, and the length of time that it persisted, stood in direct relationship with the severity of the original injury. This curious condition, which Berlin called *commotio retinæ*, was associated with some reduction of sight, episcleral congestion, and a difficulty in getting the pupil to dilate when atropine was dropped into the eye. Small retinal hemorrhages were sometimes present. Berlin explained the ophthalmoscopic picture by supposing that a rupture of the choroid was followed by bleeding and œdema of the retina. This theory has recently been opposed by Denig. That observer, as the result of experiments upon rabbits, believes that the blow upon the eyeball causes the vitreous to impinge upon the retina, to tear the internal limiting membrane, and to force the vitreous into the nerve-fiber layer. The alternate elevations and depressions thus brought about in the nerve-fiber layer of the retina are, according to Denig, the cause of the ophthalmoscopic appearances.

Since the publication of Berlin's original paper few cases of *commotio retinæ* have been recorded. Indeed, the retinal changes are of so fleeting a nature that an opportunity for observing them must occur comparatively seldom. This fact leads me to place upon record brief notes of a somewhat interesting case:

E. S., aged eleven years. First seen on July 25, 1899.

History.—At 6.45 P. M., on July 24, the patient was struck in the right eye with a cricket ball, made of cork and covered with rag cloth.

Present State.—Right eye: Small abrasion of the skin of the lower lid, with a surrounding area of redness. Some general conjunctival congestion, with a definite ecchymosis in the ocular conjunctiva, opposite the lower-outer quadrant of the cornea. Tension minus 1. The pupil distinctly sluggish and a trifle larger than the other one. Anterior chamber deep. A narrow line of blood clot lay at the bottom of the anterior chamber. V. $\frac{5}{9}$ (ii. letters). Pupil dilates imperfectly to a mydriatic. With the mirror alone some parts of the fundus oculi were seen to be unduly white. When examined more closely with the ophthalmoscope there was found a wide but defective zone of whitish fundus, situated peripherally upward, inward, and outward. No such appearances could be made out in the lower part of the fundus. The retinal vessels, which lay anterior to the affected areas, showed no changes. In most places it was possible to get beyond the whitish patches so as to see the edges of the latter. These margins were irregular, and showed white, tongue-like projections running into normal fundus. Some small islands of cloudiness lay, however, beyond the area of general haziness. Around the yellow-spot region was a white radiating appearance, but no definite white mass was present in that place. Left eye: No fundus changes. V. $\frac{5}{5}$ (iv. letters). Tension normal.

Treatment.—Vaseline to abrasion of skin of lid; atropine drops (2 grs. to the ounce—to each eye twice a day); rest in bed.

Progress.—July 26. R. V. $\frac{5}{12}$; tension still rather low. The blood clot present in anterior chamber and also on anterior capsule of lens renders it difficult to see the fundus clearly; but no white patches can be made out in the fundus.

July 27. A little blood is still present in the lower part of the anterior chamber. The parts of the retina that were milky have resumed almost their natural appearance, and the changes above mentioned are now represented merely by a faint, whitish, ill-defined stippling of the areas in question. Around the yellow spot is a system of fine radiating lines, which extend for some distance into the surrounding fundus. This is doubtless due to œdema of the retina.

July 28. R. V. $\frac{5}{6}$ (i letter); tension still slightly *minus*. Ecchymosis present in ocular conjunctiva, but the blood has disappeared from the anterior chamber. Pupil not so wide as that of the left eye, although atropine is being used to both. Faint cloudiness lower third of the cornea, made up of almost transparent dots, as may be seen with a +20 lens in certain positions of the eye. Fundus changes have disappeared; faint radiating lines, however, may still be seen around the yellow-spot region.

July 29. R. V. $\frac{5}{6}$ (ii letter) T—I. Pupil now as large as that of the other eye. Yellow-spot region still surrounded by a wide band of fine, closely set, radiating gray lines. It may be noted that the corresponding region of the left (unaffected) eye is encircled by an ordinary oval reflex.

August 1. R. V. $\frac{5}{6}$; (Tn.). A small ecchymosis still present in the ocular conjunctiva on the outer side of the cornea. No blood in anterior chamber; no corneal cloudiness. Radiating appearance still present around yellow spot of fundus.

August 9. R. V. $\frac{5}{9}$, L. V. $\frac{5}{9}$; (Tn.).

August 12. Vision unaltered. Radiating lines still present around yellow-spot region of affected eye.

September 5. The right pupil rather larger than its fellow, but no break in the continuity of the edge of the iris can be discovered to account for this. The action, both to light and to accommodation, of the pupils is equal. The radiating lines formerly present around the yellow spot of the right eye have been replaced by an ordinary oval reflex, like that present in the other fundus. Tn.; R. V. $\frac{5}{6}$ (i letter), L. V. $\frac{5}{6}$ (i letter); No. 1 Jaeger read easily.

September 7. Under atropine. R. V. = $\frac{5}{18}$ + 1.5 D. Sph. = $\frac{5}{5}$. L. V. = $\frac{5}{12}$ + 1.0 D. Sph. = $\frac{5}{5}$.

DEADY.

BOOK REVIEW.

Diseases of the Nose, Throat, and Ear. Part I. Diseases of the Nose and Throat. By S. H. Vehslage, M. D., Assistant Surgeon to the New York Ophthalmic Hospital (Throat Department). Part II. Diseases of the Ear. By G. De Wayne Hallett, M. D., Assistant Surgeon to the New York Ophthalmic Hospital. New York: Boericke & Runyon Co., 1900. Price, cloth, $3.00.

We are glad to receive this volume, for which we have been looking for quite a while; because it fills a vacancy in our homeopathic literature which has long needed filling.

It will be found a clearly and concisely written volume, very instructive and advantageous to the student and busy practitioner; it is not intended to be exhaustive enough for the specialist in these branches.

Among the many commendable points, we would call attention to thorough consideration of the subject of Diphtheria, and we are glad to say that the treatment of this disease is sufficiently broad to include antitoxin,—let it be rational, allopathic, or whatever we may denominate it. While, on the other hand, we regret not to see the appreciable doses of iodide of potash more highly recommended as antidotal to the syphilitic poison, when manifest in these localities.

The manner in which the general, topical, or mechanical and hygienic portion of the treatment is handled is to be strongly approved. The homeopathic indications are well written. Like almost all books by homeopathic authors, it shows the peculiar bent of the homeopathic physician's mind is upon the therapeutics or cure of the patient rather than upon the fine development of the pathology, ætiology, etc., of the disease.

The publishers have done well in the binding, selection of type, and leading,—making it easy to read, but we are sorry to notice a number of small typographical errors have crept in, to mar it for a somewhat critical eye.

BOOKS RECEIVED.

The following books have been received and will be reviewed in the next issue of the journal.

Sajous' "Annual and Analytical Cyclopædia of Practical Medicine." Vol. V., "Methyl-Blue to Rabies." The F. A. Davis Co.

"Diseases of the Eye," Nettleship; Sixth American from the Sixth English edition. Lea Bros. & Co.

"Injuries to the Eye in their Medico-Legal Aspect." By S. Baudry, M. D. The F. A. Davis Co.

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1. Read before the New York State Homeopathic Medical Society at the annual meeting, held in Albany, February 13 and 14, 1900.

2. Presented at the Annual Meeting of the National Society of Electro-Therapeutists, Atlantic City, N. J., September, 1900.

3. Published by the permission of Sir Frederick Hodgson, Governor of the Gold Coast Colony.

TRANSCRIBER'S NOTES

1. P. 201, changed "glycerine, aa 2 drams" to "glycerine, at 2 drams".
2. Silently corrected typographical errors.
3. Retained anachronistic, non-standard, and uncertain spellings as printed.
4. Footnotes have been re-indexed using numbers and collected together at the end of the last chapter.

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